1 $\lambda_E^{\leq}$ (Seal)

1.1 Syntax

$$
\begin{aligned}
A, B, C, R, S, T, \Gamma ::= {} & \mathsf{Int} \mid \varepsilon \mid A \to B \mid A \,\&\, B \mid \{\ell : A\} \mid \alpha_n \\
\mid {} & A \vee B \mid X_k \mid \mu T \mid \beta_i \mid \forall.\, A \\
\Delta \;\in {} & \mathbb{N} \rightharpoonup \mathsf{Typ} \qquad \emptyset \;=\; \lambda n.\ \mathsf{none} \\
e ::= {} & ? \mid e.k \mid i \mid \varepsilon \mid \lambda(A{\to}B).\, e \mid e_1 \triangleright e_2 \mid \langle e_1, \lambda(A{\to}B).\, e_2 \rangle \mid e_1\, e_2 \\
\mid {} & e_1 \,\#\, e_2 \mid \{\ell = e\} \mid e.\ell \mid (e : A) \mid \langle n \rangle\, e \mid \mathsf{seal}\ n\ R\ S\ e \mid \mathsf{unseal}\ n\ R\ S\ e \\
\mid {} & \mathsf{inl}^B\, e \mid \mathsf{inr}^A\, e \mid \mathsf{case}\ e\ \mathsf{of}\ e_1 \mid e_2 \mid \mathsf{fold}^T e \mid \mathsf{unfold}\ e \\
\mid {} & \mu\lambda(A{\to}B).\, e \mid \langle e_1, \mu\lambda(A{\to}B).\, e_2 \rangle^{B_x} \mid \Lambda_A.\, e \mid \langle e_1, \Lambda_A.\, e_2 \rangle \mid e\,[\alpha_n] \\
v ::= {} & i \mid \varepsilon \mid \langle v, \lambda(A{\to}B).\, e \rangle \mid \{\ell = v\} \mid v_1 \,\#\, v_2 \mid \langle n \rangle\, v \\
\mid {} & \mathsf{inl}^B\, v \mid \mathsf{inr}^A\, v \mid \mathsf{fold}^T v \\
\mid {} & \langle v, \mu\lambda(A{\to}B).\, e \rangle^{B_x} \mid \langle v, \Lambda_A.\, e \rangle
\end{aligned}
$$

X_k is a μ-bound type variable and β_i a $\forall$-bound *brand* variable (both de Bruijn, disjoint namespaces from each other and from concrete brands α_n); $\Lambda_A.\, e$ and its closure carry the $\forall$-body annotation A (casting at a $\forall$-type is an identity that the value must pin).

Three substitutions, all shift-free because the substitutend is closed in the relevant namespace: brand substitution $S[n{:=}R]$ (replace α_n by R), μ-unfolding $T[X_d{:=}S]^\mu$ (replace X_d by S; $d{+}1$ under μ), and $\forall\beta$-instantiation $A[\beta_d{:=}\alpha_n]$ (replace β_d by the *concrete* α_n; $d{+}1$ under $\forall$; μ binds X, not β — no bump). Each is homomorphic on all other formers; the interesting leaves:

$$
\begin{aligned}
&\alpha_m[n{:=}R] = \begin{cases} R & m = n \\ \alpha_m & m \neq n \end{cases} \qquad X_k[n{:=}R] = X_k \qquad \beta_i[n{:=}R] = \beta_i \\
&(\mu T)[n{:=}R] = \mu\, T[n{:=}R] \qquad (\forall.\, A)[n{:=}R] = \forall.\, A[n{:=}R] \\
&X_k[X_d{:=}S]^\mu = \begin{cases} S & k = d \\ X_k & k \neq d \end{cases} \qquad \beta_i[X_d{:=}S]^\mu = \beta_i \qquad \alpha_m[X_d{:=}S]^\mu = \alpha_m \\
&\beta_i[\beta_d{:=}\alpha_n] = \begin{cases} \alpha_n & i = d \\ \beta_i & i \neq d \end{cases} \qquad X_k[\beta_d{:=}\alpha_n] = X_k \\
&(\mu T)[\beta_d{:=}\alpha_n] = \mu\, T[\beta_d{:=}\alpha_n] \qquad (\forall.\, A)[\beta_d{:=}\alpha_n] = \forall.\, A[\beta_{d+1}{:=}\alpha_n]
\end{aligned}
$$

$e[\beta_d{:=}\alpha_n]$ pushes the instantiation into every type annotation of e ($d{+}1$ under Λ and its closure's annotation/body; a captured environment sits outside the binder and substitutes at d).

Occurrence predicates: $\mathsf{bv}(T)$ (some β_i occurs in T, index-blind); $\forall \in T$ (some $\forall$ occurs in T); $\mathsf{BClosed}\ c\ T$ (every β_i in T is bound below the cutoff: $i < c$ plus the enclosing binders; $c{+}1$ under $\forall$). Size measures: tsize on types and esize on terms (annotations excluded); both are preserved by $\forall\beta$-instantiation, which only trades a β_i leaf for a α_n leaf — the well-foundedness of the logical relation and of the fundamental lemma's induction rests on this.

$$
\frac{}{A\&B \ni_0 B}\ \textsc{zero} \qquad \frac{A \ni_k C}{A\&B \ni_{k+1} C}\ \textsc{succ} \qquad \frac{}{\ell \in \{\ell : A\}}\ \textsc{rcd} \qquad \frac{\ell \in A}{\ell \in A\&B}\ \textsc{andl} \qquad \frac{\ell \in B}{\ell \in A\&B}\ \textsc{andr}
$$

$$
\frac{}{\ell : A \in \{\ell : A\}}\ \textsc{zero} \qquad \frac{\ell : C \in A \qquad \neg\, \ell \in B}{\ell : C \in A\&B}\ \textsc{landl} \qquad \frac{\ell : C \in B \qquad \neg\, \ell \in A}{\ell : C \in A\&B}\ \textsc{landr}
$$

$$
\frac{}{\mathsf{Int}\ \mathsf{ord}}\ \textsc{oint} \qquad \frac{}{A \to B\ \mathsf{ord}}\ \textsc{oarr} \qquad \frac{}{\{\ell : A\}\ \mathsf{ord}}\ \textsc{orcd} \qquad \frac{}{\alpha_n\ \mathsf{ord}}\ \textsc{obrand} \qquad \frac{}{A \vee B\ \mathsf{ord}}\ \textsc{oor}
$$

$$
\frac{}{X_k\ \mathsf{ord}}\ \textsc{ovar} \qquad \frac{}{\mu T\ \mathsf{ord}}\ \textsc{omu} \qquad \frac{}{\beta_i\ \mathsf{ord}}\ \textsc{obvar} \qquad \frac{}{\forall.\, A\ \mathsf{ord}}\ \textsc{oallb} \qquad \frac{}{\rceil \varepsilon \lceil}\ \textsc{tltop} \qquad \frac{\rceil A \lceil \qquad \rceil B \lceil}{\rceil A\&B \lceil}\ \textsc{tland}
$$

$$
\frac{\rceil B \lceil}{\rceil A \to B \lceil}\ \textsc{tlarr} \qquad \frac{\rceil B \lceil}{\rceil \{\ell : B\} \lceil}\ \textsc{tlrcd}
$$

$$\mathsf{Int}^{\uparrow} = \varepsilon \quad \varepsilon^{\uparrow} = \varepsilon \quad (A \to B)^{\uparrow} = \langle \varepsilon, \lambda(A{\to}B).\, B^{\uparrow} \rangle \quad (A\&B)^{\uparrow} = A^{\uparrow} \,\#\, B^{\uparrow} \quad \{\ell : A\}^{\uparrow} = \{\ell = A^{\uparrow}\}$$
$$\alpha_n{}^{\uparrow} = (A \vee B)^{\uparrow} = X_k{}^{\uparrow} = (\mu T)^{\uparrow} = \beta_i{}^{\uparrow} = (\forall.\, A)^{\uparrow} = \varepsilon$$

1.2 Typing

Subtyping $A <: B$

$$\frac{}{\mathsf{Int} <: \mathsf{Int}}\ \textsc{sint} \qquad \frac{}{A <: \varepsilon}\ \textsc{stop} \qquad \frac{B_1 <: A_1 \qquad A_2 <: B_2}{A_1 \to A_2 <: B_1 \to B_2}\ \textsc{sarr} \qquad \frac{A_1 <: A_3}{A_1 \& A_2 <: A_3}\ \textsc{sandl}$$

$$\frac{A_2 <: A_3}{A_1 \& A_2 <: A_3}\ \textsc{sandr} \qquad \frac{A_1 <: A_2 \qquad A_1 <: A_3}{A_1 <: A_2 \& A_3}\ \textsc{sand} \qquad \frac{A <: B}{\{\ell : A\} <: \{\ell : B\}}\ \textsc{srcd} \qquad \frac{}{\alpha_n <: \alpha_n}\ \textsc{sbrand}$$

$$\frac{A_1 <: B_1 \qquad A_2 <: B_2}{A_1 \vee A_2 <: B_1 \vee B_2}\ \textsc{sor} \qquad \frac{}{X_k <: X_k}\ \textsc{svar} \qquad \frac{}{\mu T <: \mu T}\ \textsc{smu} \qquad \frac{}{\beta_i <: \beta_i}\ \textsc{sbvar} \qquad \frac{}{\forall.\, A <: \forall.\, A}\ \textsc{sallb}$$

Unions are component-wise only (no injection $A <: A \vee B$: casting would have to invent a tag); μ, X, β and $\forall$ are opaque, reflexive-only. *No* rule $\alpha_n <: R$: that would be translucency and kills representation independence.

Common ordinary supertypes and disjointness $A \sqcap B$, $A * B \;:=\; \neg(A \sqcap B)$

$$\frac{}{\mathsf{Int} \sqcap \mathsf{Int}}\ \textsc{cint} \qquad \frac{A \sqcap C}{A\&B \sqcap C}\ \textsc{candl} \qquad \frac{B \sqcap C}{A\&B \sqcap C}\ \textsc{candr} \qquad \frac{A \sqcap B}{A \sqcap B\&C}\ \textsc{crandl} \qquad \frac{A \sqcap C}{A \sqcap B\&C}\ \textsc{crandr}$$

$$\frac{B \sqcap D}{A \to B \sqcap C \to D}\ \textsc{carr} \qquad \frac{A \sqcap B}{\{\ell : A\} \sqcap \{\ell : B\}}\ \textsc{crcd} \qquad \frac{}{\alpha_n \sqcap \alpha_n}\ \textsc{cbrand} \qquad \frac{}{A \vee B \sqcap A' \vee B'}\ \textsc{coror}$$

$$\frac{}{X_k \sqcap X_k}\ \textsc{cvar} \qquad \frac{}{\mu T \sqcap \mu T}\ \textsc{cmu} \qquad \frac{}{\beta_i \sqcap \beta_j}\ \textsc{cbvar} \qquad \frac{}{\beta_i \sqcap \alpha_n}\ \textsc{cbvarbrand} \qquad \frac{}{\alpha_n \sqcap \beta_i}\ \textsc{cbrandbvar}$$

$$\frac{}{\forall.\, A \sqcap \forall.\, A}\ \textsc{callb} \qquad \frac{\mathsf{bv}(A) \vee \mathsf{bv}(A')}{\forall.\, A \sqcap \forall.\, A'}\ \textsc{callbbv} \qquad \frac{\mathsf{bv}(T) \vee \mathsf{bv}(T')}{\mu T \sqcap \mu T'}\ \textsc{cmubv}$$

The brand-variable rules are deliberately *conservative*: a β_i may be instantiated to any brand, so it costs every β_j and every α_n (unconditionally, in both directions), and two syntactically distinct $\forall$- or μ-bodies containing brand variables can become equal after instantiation (CALLBBV/CMUBV). This is exactly what makes disjointness stable under instantiation: $A * B \Rightarrow A[\beta_d{:=}\alpha_n] * B[\beta_d{:=}\alpha_n]$, with no freshness side condition anywhere (disj_bsubst). The price: nothing typed at a brand variable (or a β-containing $\mu/\forall$) can be *merged* with anything brand- or β-typed — the same wrap-it-in-records discipline unions already impose via COROR.

Casting and consistency $v \hookrightarrow_A v', \quad v_1 \approx v_2 \;:=\; \forall A\, w_1\, w_2.\; v_1 \hookrightarrow_A w_1 \Rightarrow v_2 \hookrightarrow_A w_2 \Rightarrow w_1 = w_2$

$$\frac{}{i \hookrightarrow_{\mathsf{Int}} i}\ \textsc{cint} \qquad \frac{}{v \hookrightarrow_{\varepsilon} \varepsilon}\ \textsc{ctop} \qquad \frac{\neg\rceil D\lceil \qquad C <: A \qquad B <: D}{\langle v, \lambda(A{\to}B).e\rangle \hookrightarrow_{C\to D} \langle v, \lambda(A{\to}D).e\rangle}\ \textsc{carrow}$$

$$\frac{\rceil D\lceil \qquad C <: A \qquad B <: D}{\langle v, \lambda(A{\to}B).e\rangle \hookrightarrow_{C\to D} \langle \varepsilon, \lambda(C{\to}D).D^{\uparrow}\rangle}\ \textsc{carrowtl} \qquad \frac{A\ \mathsf{ord} \qquad v_1 \hookrightarrow_A v_1'}{v_1 \# v_2 \hookrightarrow_A v_1'}\ \textsc{cmrgl}$$

$$\frac{A\ \mathsf{ord} \qquad v_2 \hookrightarrow_A v_2'}{v_1 \# v_2 \hookrightarrow_A v_2'}\ \textsc{cmrgr} \qquad \frac{v \hookrightarrow_A v_1 \qquad v \hookrightarrow_B v_2}{v \hookrightarrow_{A\&B} v_1 \# v_2}\ \textsc{cand} \qquad \frac{v \hookrightarrow_A v'}{\{\ell = v\} \hookrightarrow_{\{\ell:A\}} \{\ell = v'\}}\ \textsc{crcd}$$

$$\frac{}{\langle n\rangle\, v \hookrightarrow_{\alpha_n} \langle n\rangle\, v}\ \textsc{cwrap} \qquad \frac{v \hookrightarrow_{A'} v'}{\mathsf{inl}^{B} v \hookrightarrow_{A'\vee B'} \mathsf{inl}^{B'} v'}\ \textsc{cinl} \qquad \frac{v \hookrightarrow_{B'} v'}{\mathsf{inr}^{A} v \hookrightarrow_{A'\vee B'} \mathsf{inr}^{A'} v'}\ \textsc{cinr}$$

$$\frac{}{\mathsf{fold}^{T} v \hookrightarrow_{\mu T} \mathsf{fold}^{T} v}\ \textsc{cfold} \qquad \frac{\neg\rceil D\lceil \qquad C <: A \qquad B_x <: D}{\langle v, \mu\lambda(A{\to}B).\, e\rangle^{B_x} \hookrightarrow_{C\to D} \langle v, \mu\lambda(A{\to}B).\, e\rangle^{D}}\ \textsc{cfarrow}$$

$$\frac{\rceil D\lceil \qquad C <: A \qquad B_x <: D}{\langle v, \mu\lambda(A{\to}B).\, e\rangle^{B_x} \hookrightarrow_{C\to D} \langle \varepsilon, \lambda(C{\to}D).D^{\uparrow}\rangle}\ \textsc{cfarrowtl} \qquad \frac{}{\langle \rho, \Lambda_A.\, e\rangle \hookrightarrow_{\forall.\, A} \langle \rho, \Lambda_A.\, e\rangle}\ \textsc{cbclos}$$

Casting follows the union tag; at μ- and $\forall$-types it is the identity on folds of the same body resp. brand closures with the same annotation (the CWRAP pattern), and *undefined* at X_k and β_i (no cast rule — a term whose redex casts at a brand variable is stuck until instantiated). A fixpoint closure carries two codomains: casts rewrite only the external B_x; the internal B is pinned because beta reinstalls the self-reference at it.

Reserved-label freedom and signature well-formedness $\mathsf{NoRes}\ T, \quad \mathsf{WfSig}\ n\ R\ S$

$$\frac{}{\mathsf{NoRes\ Int}} \qquad \frac{}{\mathsf{NoRes}\ \varepsilon} \qquad \frac{}{\mathsf{NoRes}\ \alpha_n} \qquad \frac{\mathsf{NoRes}\ A \qquad \mathsf{NoRes}\ B}{\mathsf{NoRes}\ (A \to B)} \qquad \frac{\ell \neq \#\mathtt{f} \qquad \mathsf{NoRes}\ A}{\mathsf{NoRes}\ \{\ell : A\}}$$

$$\frac{\mathsf{NoRes}\ A \qquad \mathsf{NoRes}\ B}{\mathsf{NoRes}\ (A\&B)} \qquad \frac{\mathsf{NoRes}\ A \qquad \mathsf{NoRes}\ B}{\mathsf{NoRes}\ (A \vee B)} \qquad \frac{}{\mathsf{NoRes}\ X_k} \qquad \frac{\mathsf{NoRes}\ T}{\mathsf{NoRes}\ (\mu T)} \qquad \frac{}{\mathsf{NoRes}\ \beta_i} \qquad \frac{\mathsf{NoRes}\ A}{\mathsf{NoRes}\ (\forall.\, A)}$$

$$\frac{}{\mathsf{WfSig}\ n\ R\ \mathsf{Int}} \qquad \frac{}{\mathsf{WfSig}\ n\ R\ \varepsilon} \qquad \frac{}{\mathsf{WfSig}\ n\ R\ \alpha_m} \qquad \frac{\mathsf{WfSig}\ n\ R\ A \qquad \mathsf{WfSig}\ n\ R\ B}{\mathsf{WfSig}\ n\ R\ (A \to B)}$$

$$\frac{\ell \neq \#\mathtt{f} \qquad \mathsf{WfSig}\ n\ R\ A}{\mathsf{WfSig}\ n\ R\ \{\ell : A\}} \qquad \frac{\mathsf{WfSig}\ n\ R\ A \qquad \mathsf{WfSig}\ n\ R\ B \qquad A * B \qquad A[n{:=}R] * B[n{:=}R]}{\mathsf{WfSig}\ n\ R\ (A\&B)}$$

$$\frac{\mathsf{WfSig}\ n\ R\ A \qquad \mathsf{WfSig}\ n\ R\ B}{\mathsf{WfSig}\ n\ R\ (A \vee B)} \qquad \frac{\alpha_n \notin \mu T \qquad \mathsf{NoRes}\ (\mu T) \qquad \neg\mathsf{bv}(\mu T)}{\mathsf{WfSig}\ n\ R\ (\mu T)}$$

$$\frac{\alpha_n \notin \forall.\, A \qquad \mathsf{NoRes}\ (\forall.\, A) \qquad \neg\mathsf{bv}(\forall.\, A)}{\mathsf{WfSig}\ n\ R\ (\forall.\, A)}$$

WfSig has *no* X_k and no β_i case: signatures are brand-variable-free throughout (wfsig_no_hasbvar), so sealing annotations are invariant under $\forall\beta$-instantiation. μ- and $\forall$-components of a signature are opaque under sealing (the coercion is the identity on them), hence must not mention the sealed brand α_n.

Typing $\Delta; \Gamma \vdash e : A$

$$\frac{}{\Delta;\Gamma \vdash ? : \Gamma}\ \textsc{tquery} \qquad \frac{}{\Delta;\Gamma \vdash i : \mathsf{Int}}\ \textsc{tint} \qquad \frac{}{\Delta;\Gamma \vdash \varepsilon : \varepsilon}\ \textsc{tunit} \qquad \frac{\Delta;\Gamma \vdash e_1 : A \to B \qquad \Delta;\Gamma \vdash e_2 : A}{\Delta;\Gamma \vdash e_1\, e_2 : B}\ \textsc{tapp}$$

$$\frac{\Delta;\Gamma \vdash e_1 : \Gamma_1 \qquad \Delta;\Gamma_1 \vdash e_2 : A}{\Delta;\Gamma \vdash e_1 \triangleright e_2 : A}\ \textsc{tbox} \qquad \frac{\Delta;\Gamma \vdash e : B \qquad B \ni_k A}{\Delta;\Gamma \vdash e.k : A}\ \textsc{tproj} \qquad \frac{\Delta;\Gamma \vdash e : A}{\Delta;\Gamma \vdash \{\ell = e\} : \{\ell : A\}}\ \textsc{trcd}$$

$$\frac{\Delta;\Gamma \vdash e : B \qquad \ell : A \in B}{\Delta;\Gamma \vdash e.\ell : A}\ \textsc{trproj} \qquad \frac{\Delta;\Gamma \vdash e_1 : A \qquad \Delta;\Gamma\&A \vdash e_2 : B \qquad A * \Gamma \qquad A * B}{\Delta;\Gamma \vdash e_1 \# e_2 : A\&B}\ \textsc{tmrg}$$

$$\frac{v_1\ \mathsf{value} \qquad v_2\ \mathsf{value} \qquad \Delta;\varepsilon \vdash v_1 : A \qquad \Delta;\varepsilon \vdash v_2 : B \qquad A * B \qquad v_1 \approx v_2}{\Delta;\Gamma \vdash v_1 \# v_2 : A\&B}\ \textsc{tmergev}$$

$$\frac{\begin{array}{c} v_1\ \mathsf{value} \qquad v_2\ \mathsf{value} \qquad \Delta;\varepsilon \vdash v_1 : A \qquad \Delta;\varepsilon \vdash v_2 : B \\ v\ \mathsf{value} \qquad \Delta;\varepsilon \vdash v : C \qquad v \hookrightarrow_A v_1 \qquad v \hookrightarrow_B v_2 \qquad v_1 \approx v_2 \end{array}}{\Delta;\Gamma \vdash v_1 \# v_2 : A\&B}\ \textsc{tmergevc} \qquad \frac{\Gamma * A \qquad \Delta;\Gamma\&A \vdash e : B}{\Delta;\Gamma \vdash \lambda(A{\to}B).e : A \to B}\ \textsc{tlam}$$

$$\frac{v\ \mathsf{value} \qquad \Delta;\varepsilon \vdash v : \Gamma_1 \qquad \Gamma_1 * A \qquad \Delta;\Gamma_1\&A \vdash e : B' \qquad B' <: B \qquad C <: A}{\Delta;\Gamma \vdash \langle v, \lambda(A{\to}B).e\rangle : C \to B}\ \textsc{tclos}$$

$$\frac{\Delta;\Gamma \vdash e : B \qquad B <: A}{\Delta;\Gamma \vdash (e : A) : A}\ \textsc{tanno} \qquad \frac{\Delta\, n = R \qquad \neg\mathsf{bv}(R) \qquad v\ \mathsf{value} \qquad \Delta;\varepsilon \vdash v : R}{\Delta;\Gamma \vdash \langle n\rangle\, v : \alpha_n}\ \textsc{twrap}$$

$$\frac{\Delta\, n = R \qquad \mathsf{NoRes}\ R \qquad \neg\mathsf{bv}(R) \qquad \mathsf{WfSig}\ n\ R\ S \qquad \Delta;\Gamma \vdash e : S[n{:=}R]}{\Delta;\Gamma \vdash \mathsf{seal}\ n\ R\ S\ e : S}\ \textsc{tseal}$$

$$\frac{\Delta\, n = R \qquad \mathsf{NoRes}\ R \qquad \neg\mathsf{bv}(R) \qquad \mathsf{WfSig}\ n\ R\ S \qquad \Delta;\Gamma \vdash e : S}{\Delta;\Gamma \vdash \mathsf{unseal}\ n\ R\ S\ e : S[n{:=}R]}\ \textsc{tunseal} \qquad \frac{\Delta;\Gamma \vdash e : A}{\Delta;\Gamma \vdash \mathsf{inl}^B e : A \vee B}\ \textsc{tinl}$$

$$\frac{\Delta;\Gamma \vdash e : B}{\Delta;\Gamma \vdash \mathsf{inr}^A e : A \vee B}\ \textsc{tinr}$$

$$\frac{\Delta;\Gamma \vdash e : A \vee B \qquad \Gamma * A \qquad \Gamma * B \qquad \Delta;\Gamma\&A \vdash e_1 : C \qquad \Delta;\Gamma\&B \vdash e_2 : C}{\Delta;\Gamma \vdash \mathsf{case}\ e\ \mathsf{of}\ e_1 \mid e_2 : C}\ \textsc{tcase}$$

$$\frac{\neg\forall \in T \qquad \Delta;\Gamma \vdash e : T[X_0{:=}\mu T]^\mu}{\Delta;\Gamma \vdash \mathsf{fold}^T e : \mu T}\ \textsc{tfold} \qquad \frac{\neg\forall \in T \qquad \Delta;\Gamma \vdash e : \mu T \qquad A = T[X_0{:=}\mu T]^\mu}{\Delta;\Gamma \vdash \mathsf{unfold}\ e : A}\ \textsc{tunfold}$$

$$\frac{\Gamma * (A{\to}B) \qquad (\Gamma\&(A{\to}B)) * A \qquad \Delta;(\Gamma\&(A{\to}B))\&A \vdash e : B}{\Delta;\Gamma \vdash \mu\lambda(A{\to}B).e : A \to B}\ \textsc{tflam}$$

$$\frac{\begin{array}{c} v\ \mathsf{value} \qquad \Delta;\varepsilon \vdash v : \Gamma_1 \qquad \Gamma_1 * (A{\to}B) \qquad (\Gamma_1\&(A{\to}B)) * A \\ \Delta;(\Gamma_1\&(A{\to}B))\&A \vdash e : B' \qquad B' <: B \qquad B <: B_x \qquad C <: A \end{array}}{\Delta;\Gamma \vdash \langle v, \mu\lambda(A{\to}B).\, e\rangle^{B_x} : C \to B_x}\ \textsc{tfclos} \qquad \frac{\neg\mathsf{bv}(\Gamma) \qquad \Delta;\Gamma \vdash e : A}{\Delta;\Gamma \vdash \Lambda_A.\, e : \forall.\, A}\ \textsc{tblam}$$

$$\frac{\rho\ \mathsf{value} \qquad \Delta;\varepsilon \vdash \rho : \Gamma_1 \qquad \neg\mathsf{bv}(\Gamma_1) \qquad \Delta;\Gamma_1 \vdash e : A}{\Delta;\Gamma \vdash \langle \rho, \Lambda_A.\, e\rangle : \forall.\, A}\ \textsc{tbclos}$$

$$\frac{\Delta;\Gamma \vdash e : \forall.\, A \qquad \mathsf{BClosed}\ 1\ A \qquad T = A[\beta_0{:=}\alpha_n]}{\Delta;\Gamma \vdash e\,[\alpha_n] : T}\ \textsc{tbapp}$$

TMERGEV is the module-composition merge (disjoint components); TMERGEVC is the cast-duplicated merge

produced by casting one value at $A \,\&\, B$, recording the common origin v — that record is what keeps the rule stable under $\forall\beta$-instantiation (consistency of the substituted components is rebuilt from casts of the substituted origin rather than assumed). The three $\forall$-fences: $\neg\forall{\in}T$ on TFOLD/TUNFOLD (μ-unfolding does not shift brand variables); $\mathsf{BClosed}\ 1\ A$ on TBAPP (the shift-free beta is index-coherent only for single-binder bodies); $\neg\mathsf{bv}(\cdot)$ on captured contexts and representations (they must be instantiation-invariant).

$$\Delta \sqsubseteq \Delta' \;:=\; \forall n\, R.\ \Delta\, n = R \Rightarrow \Delta'\, n = R$$

1.3 Semantics

Value lookups $\quad v \ni_k v', \quad \ell : v' \in v$

$$\frac{}{v_1 \# v_2 \ni_0 v_2}\ \text{LVZERO} \qquad \frac{v_1 \ni_k v_3}{v_1 \# v_2 \ni_{k+1} v_3}\ \text{LVSUCC} \qquad \frac{}{\ell : e \in \{\ell = e\}}\ \text{RVLZERO} \qquad \frac{\ell : e \in e_1}{\ell : e \in e_1 \# e_2}\ \text{VLANDL}$$

$$\frac{\ell : e \in e_2}{\ell : e \in e_1 \# e_2}\ \text{VLANDR}$$

Sealing coercions $\quad v \hookrightarrow_S^{\mathsf{seal}\, n\, R} w, \quad v \hookrightarrow_S^{\mathsf{unseal}\, n\, R} w \qquad (\mathsf{pxe}(c) := \varepsilon \# \{\#\mathtt{f} = c\}, \quad \mathsf{pxf} := (?.1).\#\mathtt{f})$

$$\frac{}{v \hookrightarrow_{\alpha_n}^{\mathsf{seal}\, n\, R} \langle n \rangle\, v}\ \text{BRAND-EQ} \qquad \frac{m \neq n}{v \hookrightarrow_{\alpha_m}^{\mathsf{seal}\, n\, R} v}\ \text{BRAND-NE} \qquad \frac{}{i \hookrightarrow_{\mathsf{Int}}^{\mathsf{seal}\, n\, R} i}\ \text{INT} \qquad \frac{}{v \hookrightarrow_{\varepsilon}^{\mathsf{seal}\, n\, R} \varepsilon}\ \text{TOP}$$

$$\frac{v_1 \hookrightarrow_A^{\mathsf{seal}\, n\, R} w_1 \qquad v_2 \hookrightarrow_B^{\mathsf{seal}\, n\, R} w_2}{v_1 \# v_2 \hookrightarrow_{A\&B}^{\mathsf{seal}\, n\, R} w_1 \# w_2}\ \text{AND} \qquad \frac{v \hookrightarrow_A^{\mathsf{seal}\, n\, R} w}{\{\ell = v\} \hookrightarrow_{\{\ell:A\}}^{\mathsf{seal}\, n\, R} \{\ell = w\}}\ \text{RCD}$$

$$\frac{}{c \hookrightarrow_{A\to B}^{\mathsf{seal}\, n\, R} \langle \mathsf{pxe}(c), \lambda(A{\to}B).\ \mathsf{seal}\ n\ R\ B\ (\mathsf{pxf}\ (\mathsf{unseal}\ n\ R\ A\ ?.0))\rangle}\ \text{ARR} \qquad \frac{v \hookrightarrow_A^{\mathsf{seal}\, n\, R} w}{\mathsf{inl}^{B[n:=R]} v \hookrightarrow_{A\vee B}^{\mathsf{seal}\, n\, R} \mathsf{inl}^{B} w}\ \text{INL}$$

$$\frac{v \hookrightarrow_B^{\mathsf{seal}\, n\, R} w}{\mathsf{inr}^{A[n:=R]} v \hookrightarrow_{A\vee B}^{\mathsf{seal}\, n\, R} \mathsf{inr}^{A} w}\ \text{INR} \qquad \frac{}{v \hookrightarrow_{X_k}^{\mathsf{seal}\, n\, R} v}\ \text{VAR} \qquad \frac{}{v \hookrightarrow_{\mu T}^{\mathsf{seal}\, n\, R} v}\ \text{MU} \qquad \frac{}{v \hookrightarrow_{\beta_i}^{\mathsf{seal}\, n\, R} v}\ \text{BVAR}$$

$$\frac{}{v \hookrightarrow_{\forall.\, A}^{\mathsf{seal}\, n\, R} v}\ \text{ALLB}$$

$$\frac{}{\langle n \rangle\, v \hookrightarrow_{\alpha_n}^{\mathsf{unseal}\, n\, R} v}\ \text{BRAND-EQ} \qquad \frac{m \neq n}{v \hookrightarrow_{\alpha_m}^{\mathsf{unseal}\, n\, R} v}\ \text{BRAND-NE} \qquad \frac{}{i \hookrightarrow_{\mathsf{Int}}^{\mathsf{unseal}\, n\, R} i}\ \text{INT} \qquad \frac{}{v \hookrightarrow_{\varepsilon}^{\mathsf{unseal}\, n\, R} \varepsilon}\ \text{TOP}$$

$$\frac{v_1 \hookrightarrow_A^{\mathsf{unseal}\, n\, R} w_1 \qquad v_2 \hookrightarrow_B^{\mathsf{unseal}\, n\, R} w_2}{v_1 \# v_2 \hookrightarrow_{A\&B}^{\mathsf{unseal}\, n\, R} w_1 \# w_2}\ \text{AND} \qquad \frac{v \hookrightarrow_A^{\mathsf{unseal}\, n\, R} w}{\{\ell = v\} \hookrightarrow_{\{\ell:A\}}^{\mathsf{unseal}\, n\, R} \{\ell = w\}}\ \text{RCD}$$

$$\frac{}{c \hookrightarrow_{A\to B}^{\mathsf{unseal}\, n\, R} \langle \mathsf{pxe}(c), \lambda(A[n:=R]{\to}B[n:=R]).\ \mathsf{unseal}\ n\ R\ B\ (\mathsf{pxf}\ (\mathsf{seal}\ n\ R\ A\ ?.0))\rangle}\ \text{ARR}$$

$$\frac{v \hookrightarrow_A^{\mathsf{unseal}\, n\, R} w}{\mathsf{inl}^{B} v \hookrightarrow_{A\vee B}^{\mathsf{unseal}\, n\, R} \mathsf{inl}^{B[n:=R]} w}\ \text{INL} \qquad \frac{v \hookrightarrow_B^{\mathsf{unseal}\, n\, R} w}{\mathsf{inr}^{A} v \hookrightarrow_{A\vee B}^{\mathsf{unseal}\, n\, R} \mathsf{inr}^{A[n:=R]} w}\ \text{INR} \qquad \frac{}{v \hookrightarrow_{X_k}^{\mathsf{unseal}\, n\, R} v}\ \text{VAR}$$

$$\frac{}{v \hookrightarrow_{\mu T}^{\mathsf{unseal}\, n\, R} v}\ \text{MU} \qquad \frac{}{v \hookrightarrow_{\beta_i}^{\mathsf{unseal}\, n\, R} v}\ \text{BVAR} \qquad \frac{}{v \hookrightarrow_{\forall.\, A}^{\mathsf{unseal}\, n\, R} v}\ \text{ALLB}$$

The coercions are the identity at $X/\mu/\beta/\forall$ (opaque under sealing; type-correct because WfSig forbids α_n and brand variables there).

Small-step $v \vdash e \hookrightarrow e'$ **(all rules carry** v value**; value premises shown)**

$$\frac{}{v \vdash ? \hookrightarrow v}\ \textsc{squery} \qquad \frac{v \vdash e_1 \hookrightarrow e_1'}{v \vdash e_1\, e_2 \hookrightarrow e_1'\, e_2}\ \textsc{sappl} \qquad \frac{v_1\ \mathsf{value} \qquad v \vdash e_2 \hookrightarrow e_2'}{v \vdash v_1\, e_2 \hookrightarrow v_1\, e_2'}\ \textsc{sappr}$$

$$\frac{v_1\ \mathsf{value} \qquad v_2\ \mathsf{value} \qquad v_2 \hookrightarrow_A v_2'}{v \vdash \langle v_1, \lambda(A{\to}B).e\rangle\, v_2 \hookrightarrow (v_1 \mathbin{\#} v_2') \triangleright (e : B)}\ \textsc{sbeta} \qquad \frac{}{v \vdash \lambda(A{\to}B).e \hookrightarrow \langle v, \lambda(A{\to}B).e\rangle}\ \textsc{sclos}$$

$$\frac{v \vdash e_1 \hookrightarrow e_1'}{v \vdash e_1 \triangleright e_2 \hookrightarrow e_1' \triangleright e_2}\ \textsc{sboxl} \qquad \frac{v_1\ \mathsf{value} \qquad v_1 \vdash e_2 \hookrightarrow e_2'}{v \vdash v_1 \triangleright e_2 \hookrightarrow v_1 \triangleright e_2'}\ \textsc{sboxr} \qquad \frac{v_1\ \mathsf{value} \qquad v_2\ \mathsf{value}}{v \vdash v_1 \triangleright v_2 \hookrightarrow v_2}\ \textsc{sboxv}$$

$$\frac{v \vdash e_1 \hookrightarrow e_1'}{v \vdash e_1 \mathbin{\#} e_2 \hookrightarrow e_1' \mathbin{\#} e_2}\ \textsc{smrgl} \qquad \frac{v_1\ \mathsf{value} \qquad v \mathbin{\#} v_1 \vdash e_2 \hookrightarrow e_2'}{v \vdash v_1 \mathbin{\#} e_2 \hookrightarrow v_1 \mathbin{\#} e_2'}\ \textsc{smrgr} \qquad \frac{v \vdash e \hookrightarrow e'}{v \vdash e.k \hookrightarrow e'.k}\ \textsc{sproj}$$

$$\frac{v_1\ \mathsf{value} \qquad v_1 \ni_k v'}{v \vdash v_1.k \hookrightarrow v'}\ \textsc{sprojv} \qquad \frac{v \vdash e \hookrightarrow e'}{v \vdash \{\ell = e\} \hookrightarrow \{\ell = e'\}}\ \textsc{slrec} \qquad \frac{v \vdash e \hookrightarrow e'}{v \vdash e.\ell \hookrightarrow e'.\ell}\ \textsc{srproj}$$

$$\frac{v_1\ \mathsf{value} \qquad \ell : v' \in v_1}{v \vdash v_1.\ell \hookrightarrow v'}\ \textsc{srprojv} \qquad \frac{v \vdash e \hookrightarrow e'}{v \vdash (e : A) \hookrightarrow (e' : A)}\ \textsc{sanno} \qquad \frac{v_1\ \mathsf{value} \qquad v_1 \hookrightarrow_A v'}{v \vdash (v_1 : A) \hookrightarrow v'}\ \textsc{sannov}$$

$$\frac{v \vdash e \hookrightarrow e'}{v \vdash \langle n\rangle\, e \hookrightarrow \langle n\rangle\, e'}\ \textsc{swrap} \qquad \frac{v \vdash e \hookrightarrow e'}{v \vdash \mathsf{seal}\ n\ R\ S\ e \hookrightarrow \mathsf{seal}\ n\ R\ S\ e'}\ \textsc{sseal} \qquad \frac{v_1\ \mathsf{value} \qquad v_1 \hookrightarrow_S^{\mathsf{seal}\, n\, R} w}{v \vdash \mathsf{seal}\ n\ R\ S\ v_1 \hookrightarrow w}\ \textsc{sealv}$$

$$\frac{v \vdash e \hookrightarrow e'}{v \vdash \mathsf{unseal}\ n\ R\ S\ e \hookrightarrow \mathsf{unseal}\ n\ R\ S\ e'}\ \textsc{sunseal} \qquad \frac{v_1\ \mathsf{value} \qquad v_1 \hookrightarrow_S^{\mathsf{unseal}\, n\, R} w}{v \vdash \mathsf{unseal}\ n\ R\ S\ v_1 \hookrightarrow w}\ \textsc{sunsealv}$$

$$\frac{v \vdash e \hookrightarrow e'}{v \vdash \mathsf{inl}^B e \hookrightarrow \mathsf{inl}^B e'}\ \textsc{sinl} \qquad \frac{v \vdash e \hookrightarrow e'}{v \vdash \mathsf{inr}^A e \hookrightarrow \mathsf{inr}^A e'}\ \textsc{sinr} \qquad \frac{v \vdash e \hookrightarrow e'}{v \vdash \mathsf{case}\ e\ \mathsf{of}\ e_1 \mid e_2 \hookrightarrow \mathsf{case}\ e'\ \mathsf{of}\ e_1 \mid e_2}\ \textsc{scase}$$

$$\frac{v_1\ \mathsf{value}}{v \vdash \mathsf{case}\ (\mathsf{inl}^B v_1)\ \mathsf{of}\ e_1 \mid e_2 \hookrightarrow (v \mathbin{\#} v_1) \triangleright e_1}\ \textsc{scasel} \qquad \frac{v_1\ \mathsf{value}}{v \vdash \mathsf{case}\ (\mathsf{inr}^A v_1)\ \mathsf{of}\ e_1 \mid e_2 \hookrightarrow (v \mathbin{\#} v_1) \triangleright e_2}\ \textsc{scaser}$$

$$\frac{v \vdash e \hookrightarrow e'}{v \vdash \mathsf{fold}^T e \hookrightarrow \mathsf{fold}^T e'}\ \textsc{sfold} \qquad \frac{v \vdash e \hookrightarrow e'}{v \vdash \mathsf{unfold}\ e \hookrightarrow \mathsf{unfold}\ e'}\ \textsc{sunfold} \qquad \frac{v_1\ \mathsf{value}}{v \vdash \mathsf{unfold}\ (\mathsf{fold}^T v_1) \hookrightarrow v_1}\ \textsc{sunfoldv}$$

$$\frac{}{v \vdash \mu\lambda(A{\to}B).e \hookrightarrow \langle v, \mu\lambda(A{\to}B).\, e\rangle^B}\ \textsc{sflam}$$

$$\frac{v_1\ \mathsf{value} \qquad v_2\ \mathsf{value} \qquad v_2 \hookrightarrow_A v_2'}{v \vdash \langle v_1, \mu\lambda(A{\to}B).\, e\rangle^{B_x}\, v_2 \hookrightarrow \big((v_1 \mathbin{\#} \langle v_1, \mu\lambda(A{\to}B).\, e\rangle^B) \mathbin{\#} v_2'\big) \triangleright (e : B_x)}\ \textsc{sfbeta}$$

$$\frac{}{v \vdash \Lambda_A.\, e \hookrightarrow \langle v, \Lambda_A.\, e\rangle}\ \textsc{sblam} \qquad \frac{v \vdash e \hookrightarrow e'}{v \vdash e\,[\alpha_n] \hookrightarrow e'\,[\alpha_n]}\ \textsc{sbappl} \qquad \frac{\rho\ \mathsf{value}}{v \vdash \langle \rho, \Lambda_A.\, e\rangle\,[\alpha_n] \hookrightarrow \rho \triangleright e[\beta_0{:=}\alpha_n]}\ \textsc{sbappbeta}$$

Fixpoint beta reinstalls the self-reference with the external codomain reset to the internal B and reseals the result at the client-facing B_x. $\forall$-beta is pure annotation rewriting: no argument cast, no reseal — the instantiated body runs under the captured environment.

$$\frac{}{v \vdash e \hookrightarrow^* e}\ \textsc{refl} \qquad \frac{v \vdash e_1 \hookrightarrow e_2 \qquad v \vdash e_2 \hookrightarrow^* e_3}{v \vdash e_1 \hookrightarrow^* e_3}\ \textsc{step}$$

1.4 Meta-theory

Theorem 1.1 (gdeterminism)**.**

$$\Delta;\Gamma \vdash e : A \ \wedge\ \Delta;\varepsilon \vdash v : \Gamma \ \wedge\ v \vdash e \hookrightarrow e_1 \ \wedge\ v \vdash e \hookrightarrow e_2 \implies e_1 = e_2$$

Theorem 1.2 (gprogress)**.**

$$\Delta;\Gamma \vdash e : A \ \wedge\ v\ \mathsf{value} \ \wedge\ \Delta;\varepsilon \vdash v : \Gamma \implies e\ \mathsf{value} \ \vee\ \exists e'.\ v \vdash e \hookrightarrow e'$$

Theorem 1.3 (gpreservation)**.**

$$v \vdash e \hookrightarrow e' \ \wedge\ \Delta;\Gamma \vdash e : A \ \wedge\ \Delta;\varepsilon \vdash v : \Gamma \implies \Delta;\Gamma \vdash e' : A$$

Theorem 1.4 (hastype_weaken_store)**.**

$$\Delta \sqsubseteq \Delta' \ \wedge\ \Delta;\Gamma \vdash e : A \implies \Delta';\Gamma \vdash e : A$$

Definition 1.5 (viewL, viewR, WrapRel, RawRel, LRg, LR)**.** $o \in \{\mathsf{none}\} \cup \mathbb{N} \times \mathsf{Typ} \times \mathsf{Typ}$*;* $\eta : \mathbb{N} \to \mathsf{Exp} \to \mathsf{Exp} \to \mathsf{Prop}$*;* *stores* Δ_1, Δ_2 *ambient.*

$$\begin{array}{ll} \mathsf{viewL}\ \mathsf{none}\, T = T & \mathsf{viewL}\ (n, R_1, R_2)\, T = T[n{:=}R_1] \\ \mathsf{viewR}\ \mathsf{none}\, T = T & \mathsf{viewR}\ (n, R_1, R_2)\, T = T[n{:=}R_2] \end{array}$$

$$\begin{aligned} \mathsf{WrapRel}\ m\ (v_1, v_2) &= \Delta_1;\varepsilon \vdash v_1 : \alpha_m \wedge \Delta_2;\varepsilon \vdash v_2 : \alpha_m \wedge \exists w_1\, w_2.\ v_1 = \langle m \rangle\, w_1 \wedge v_2 = \langle m \rangle\, w_2 \wedge \eta\, m\, w_1\, w_2 \\ \mathsf{RawRel}\ n\ R_1\ R_2\ (v_1, v_2) &= v_1\ \mathsf{value} \wedge v_2\ \mathsf{value} \wedge \Delta_1;\varepsilon \vdash v_1 : R_1 \wedge \Delta_2;\varepsilon \vdash v_2 : R_2 \wedge \eta\, n\, v_1\, v_2 \end{aligned}$$

$$\begin{aligned} \mathcal{V}[\![\mathsf{Int}]\!]^o(v_1, v_2) &= \exists i.\ v_1 = i \wedge v_2 = i \\ \mathcal{V}[\![\varepsilon]\!]^o(v_1, v_2) &= v_1 = \varepsilon \wedge v_2 = \varepsilon \\ \mathcal{V}[\![\alpha_m]\!]^o(v_1, v_2) &= \begin{cases} \mathsf{RawRel}\ n\ R_1\ R_2\ (v_1, v_2) & o = (n, R_1, R_2),\ m = n \\ \mathsf{WrapRel}\ m\ (v_1, v_2) & \textit{otherwise} \end{cases} \\ \mathcal{V}[\![A \& B]\!]^o(v_1, v_2) &= \Delta_1;\varepsilon \vdash v_1 : \mathsf{viewL}\, o\, (A \& B) \wedge \Delta_2;\varepsilon \vdash v_2 : \mathsf{viewR}\, o\, (A \& B) \wedge \\ &\quad \exists a_1\, b_1\, a_2\, b_2.\ v_1 = a_1 \,\#\, b_1 \wedge v_2 = a_2 \,\#\, b_2 \wedge \mathcal{V}[\![A]\!]^o(a_1, a_2) \wedge \mathcal{V}[\![B]\!]^o(b_1, b_2) \\ \mathcal{V}[\![\{\ell : A\}]\!]^o(v_1, v_2) &= \exists w_1\, w_2.\ v_1 = \{\ell = w_1\} \wedge v_2 = \{\ell = w_2\} \wedge \mathcal{V}[\![A]\!]^o(w_1, w_2) \\ \mathcal{V}[\![A \to B]\!]^o(v_1, v_2) &= v_1\ \mathsf{value} \wedge v_2\ \mathsf{value} \wedge \Delta_1;\varepsilon \vdash v_1 : \mathsf{viewL}\, o\, (A{\to}B) \wedge \Delta_2;\varepsilon \vdash v_2 : \mathsf{viewR}\, o\, (A{\to}B) \wedge \\ &\quad \forall u_1\, u_2.\ \mathcal{V}[\![A]\!]^o(u_1, u_2) \Rightarrow \forall \rho_1\, \rho_2.\ \rho_1\ \mathsf{value} \Rightarrow \rho_2\ \mathsf{value} \Rightarrow \\ &\quad\quad \exists w_1\, w_2.\ \rho_1 \vdash v_1\, u_1 \hookrightarrow^* w_1 \wedge \rho_2 \vdash v_2\, u_2 \hookrightarrow^* w_2 \wedge \mathcal{V}[\![B]\!]^o(w_1, w_2) \end{aligned}$$

$$\begin{aligned} \mathcal{V}[\![A \vee B]\!]^o(v_1, v_2) &= \big(\exists w_1\, w_2.\ v_1 = \mathsf{inl}^{\mathsf{viewL}\, o\, B} w_1 \wedge v_2 = \mathsf{inl}^{\mathsf{viewR}\, o\, B} w_2 \wedge \mathcal{V}[\![A]\!]^o(w_1, w_2)\big) \vee \\ &\quad \big(\exists w_1\, w_2.\ v_1 = \mathsf{inr}^{\mathsf{viewL}\, o\, A} w_1 \wedge v_2 = \mathsf{inr}^{\mathsf{viewR}\, o\, A} w_2 \wedge \mathcal{V}[\![B]\!]^o(w_1, w_2)\big) \\ \mathcal{V}[\![X_k]\!]^o(v_1, v_2) &= \bot \qquad\qquad \mathcal{V}[\![\beta_i]\!]^o(v_1, v_2) \ =\ \bot \\ \mathcal{V}[\![\mu T]\!]^o(v_1, v_2) &= v_1\ \mathsf{value} \wedge v_2\ \mathsf{value} \wedge \Delta_1;\varepsilon \vdash v_1 : \mathsf{viewL}\, o\, (\mu T) \wedge \Delta_2;\varepsilon \vdash v_2 : \mathsf{viewR}\, o\, (\mu T) \wedge v_1 = v_2 \\ \mathcal{V}[\![\forall.\, A]\!]^o(v_1, v_2) &= v_1\ \mathsf{value} \wedge v_2\ \mathsf{value} \wedge \Delta_1;\varepsilon \vdash v_1 : \mathsf{viewL}\, o\, (\forall.\, A) \wedge \Delta_2;\varepsilon \vdash v_2 : \mathsf{viewR}\, o\, (\forall.\, A) \wedge \\ &\quad \forall m.\ \forall \rho_1\, \rho_2.\ \rho_1\ \mathsf{value} \Rightarrow \rho_2\ \mathsf{value} \Rightarrow \\ &\quad\quad \exists w_1\, w_2.\ \rho_1 \vdash v_1\, [\alpha_m] \hookrightarrow^* w_1 \wedge \rho_2 \vdash v_2\, [\alpha_m] \hookrightarrow^* w_2 \wedge \mathcal{V}[\![A[\beta_0{:=}\alpha_m]]\!]^o(w_1, w_2) \end{aligned}$$

$$\mathcal{V}[\![T]\!] \ :=\ \mathcal{V}[\![T]\!]^{\mathsf{none}}$$

The $\forall$*-clause is* instantiation closure*: related generic values yield, at every concrete brand* m*, related runs of the instantiation at the substituted body type.* η *is never extended and no freshness is demanded — at brand-*m *positions the relation is whatever* η *already says about* m*, which is exactly what representation independence needs (the client instantiates at the provider's own allocated brand, where* η *relates the two representations). Typings stay at the* syntactic *type, so degenerate* β*-annotated values relate vacuously through the* $\bot$ *clause at* β_i*. The recursion descends on* tsize *(*$A[\beta_0{:=}\alpha_m]$ *is not a subterm but is size-preserving), so the definition is by well-founded recursion.* μ *stays opaque (a finer clause needs step indexing); free theorems over* arbitrary *relations at* β *(relational parametricity proper) are future work.*

Definition 1.6 (Finitary)**.** Finitary *e: the least predicate closed under every term former* except *the fixpoints* $\mu\lambda(A{\to}B).e$ *and their closures. Generic terms* $\Lambda/\langle\cdot, \Lambda\rangle/\cdot\,[\alpha_n]$ are *included:* $\forall$*-beta rewrites only annotations and never duplicates subterms, so the fragment stays normalizing. The relational layer below is stated for this fragment; the safety layer (determinism, progress, preservation) covers the full language.*

Definition 1.7 (OpenAdm, OpenBrand, SemTyp)**.**

$$\mathsf{OpenAdm}\ \eta\ o = \forall n\, R_1\, R_2.\ o = (n, R_1, R_2) \Rightarrow \neg\rceil R_1 \lceil \wedge \neg\rceil R_2 \lceil \wedge \\ (\forall w_1\, w_2.\ \eta\, n\, w_1\, w_2 \Rightarrow \forall w_1'\, w_2'.\ w_1 \hookrightarrow_{R_1} w_1' \Rightarrow w_2 \hookrightarrow_{R_2} w_2' \Rightarrow \eta\, n\, w_1'\, w_2')$$

$$\mathsf{OpenBrand}\ \Delta_1\ \Delta_2\ \eta\ n\ R_1\ R_2\ =\ \begin{cases} \Delta_1\, n = R_1\ \wedge\ \Delta_2\, n = R_2\ \wedge\ \mathsf{NoRes}\ R_1\ \wedge\ \mathsf{NoRes}\ R_2\ \wedge\ \neg\rceil R_1 \lceil \wedge \neg\rceil R_2 \lceil \wedge \\ \forall w_1\, w_2.\ \eta\, n\, w_1\, w_2 \Rightarrow \forall w_1'\, w_2'.\ w_1 \hookrightarrow_{R_1} w_1' \Rightarrow w_2 \hookrightarrow_{R_2} w_2' \Rightarrow \eta\, n\, w_1'\, w_2' \end{cases}$$

$$\mathsf{SemTyp}\ \Delta_1\ \Delta_2\ \eta\ \Gamma\ e_1\ e_2\ A\ =\ \forall \rho_1\, \rho_2.\ \mathcal{V}[\![\Gamma]\!](\rho_1, \rho_2) \Rightarrow \exists w_1\, w_2.\ \rho_1 \vdash e_1 \hookrightarrow^* w_1 \wedge \rho_2 \vdash e_2 \hookrightarrow^* w_2 \wedge \mathcal{V}[\![A]\!](w_1, w_2)$$

Theorem 1.8 (fundamental)**.**

$$\emptyset; \Gamma \vdash e : A\ \wedge\ \mathsf{Finitary}\ e\ \Longrightarrow\ \mathsf{SemTyp}\ \Delta_1\ \Delta_2\ \eta\ \Gamma\ e\ e\ A$$

Proved by strong induction on size *e: the Λ-cases recurse at the* instantiated *body — the unsubstituted body cannot run (casts at β annotations are stuck) — whose typing comes from the instantiation lemma $\Delta; \Gamma \vdash e : A \Rightarrow \Delta; \Gamma[\beta_d{:=}\alpha_n] \vdash e[\beta_d{:=}\alpha_n] : A[\beta_d{:=}\alpha_n]$ (hastype_bsubst), not from a subderivation.*

Theorem 1.9 (normalization)**.**

$$\emptyset; \varepsilon \vdash e : A\ \wedge\ \mathsf{Finitary}\ e\ \Longrightarrow\ \exists v.\ v\ \mathsf{value}\ \wedge\ \varepsilon \vdash e \hookrightarrow^* v$$

Theorem 1.10 (sealing)**.**

$$\mathcal{V}[\![A]\!](p_1', p_2')\ \wedge\ \emptyset; A \vdash e : \mathsf{Int}\ \wedge\ \mathsf{Finitary}\ e\ \Longrightarrow\ \exists i.\ p_1' \vdash e \hookrightarrow^* i\ \wedge\ p_2' \vdash e \hookrightarrow^* i$$

Theorem 1.11 (coe_lr)**.**

$$\mathsf{OpenBrand}\ \Delta_1\ \Delta_2\ \eta\ n\ R_1\ R_2\ \wedge\ \mathsf{WfSig}\ n\ R_1\ S\ \wedge\ \mathsf{WfSig}\ n\ R_2\ S\ \Longrightarrow$$

$$\big(\forall v_1\, v_2\, w_1\, w_2.\ \mathcal{V}[\![S]\!]^{(n,R_1,R_2)}(v_1, v_2) \Rightarrow v_1 \hookrightarrow_S^{\mathsf{seal}\, n\, R_1} w_1 \Rightarrow v_2 \hookrightarrow_S^{\mathsf{seal}\, n\, R_2} w_2 \Rightarrow \mathcal{V}[\![S]\!](w_1, w_2)\big) \wedge \\ \big(\forall v_1\, v_2\, w_1\, w_2.\ \mathcal{V}[\![S]\!](v_1, v_2) \Rightarrow v_1 \hookrightarrow_S^{\mathsf{unseal}\, n\, R_1} w_1 \Rightarrow v_2 \hookrightarrow_S^{\mathsf{unseal}\, n\, R_2} w_2 \Rightarrow \mathcal{V}[\![S]\!]^{(n,R_1,R_2)}(w_1, w_2)\big)$$

Theorem 1.12 (representation_independence)**.**

$$\mathsf{OpenBrand}\ \Delta_1\ \Delta_2\ \eta\ n\ R_1\ R_2\ \wedge\ \mathsf{WfSig}\ n\ R_1\ S\ \wedge\ \mathsf{WfSig}\ n\ R_2\ S\ \wedge \\ \mathcal{V}[\![S]\!]^{(n,R_1,R_2)}(p_1, p_2)\ \wedge\ \emptyset; S \vdash e : \mathsf{Int}\ \wedge\ \mathsf{Finitary}\ e \\ \Longrightarrow\ \exists i.\ \varepsilon \vdash (\mathsf{seal}\ n\ R_1\ S\ p_1) \triangleright e \hookrightarrow^* i\ \wedge\ \varepsilon \vdash (\mathsf{seal}\ n\ R_2\ S\ p_2) \triangleright e \hookrightarrow^* i$$

Theorem 1.13 (generic_client_representation_independence)**.** *One* generic *client, typed once at $\forall.\,(S_\beta \to \mathsf{Int})$ with the signature abstract in the brand it will be instantiated at, observes the same integer against two sealed providers of that signature over different representations. Writing $S^* := S_\beta[\beta_0{:=}\alpha_n]$:*

$$\mathsf{OpenBrand}\ \Delta_1\ \Delta_2\ \eta\ n\ R_1\ R_2\ \wedge\ \mathsf{WfSig}\ n\ R_1\ S^*\ \wedge\ \mathsf{WfSig}\ n\ R_2\ S^*\ \wedge \\ \mathcal{V}[\![S^*]\!]^{(n,R_1,R_2)}(p_1, p_2)\ \wedge\ \emptyset; \varepsilon \vdash c_t : \forall.\,(S_\beta \to \mathsf{Int})\ \wedge\ \mathsf{Finitary}\ c_t \\ \Longrightarrow\ \exists i.\ \varepsilon \vdash (c_t\,[\alpha_n])\,(\mathsf{seal}\ n\ R_1\ S^*\ p_1) \hookrightarrow^* i\ \wedge\ \varepsilon \vdash (c_t\,[\alpha_n])\,(\mathsf{seal}\ n\ R_2\ S^*\ p_2) \hookrightarrow^* i$$

The client instantiates at the provider's own *allocated brand n: the $\forall$-clause needs no freshness, and η at n is exactly the relation between the two representations, so $\cdot\,[\alpha_n]$ lands the client in the concrete-signature world where representation_independence already lives. Both providers seal at the* same *brand n — the two representations live in the two stores, as in every RI statement ($\Delta\, n$ is functional). A single client facing providers allocated at* different *brands is the brand-renaming theorem (§3), not this one. What this layer adds beyond representation_independence is (i) coverage:* Finitary *now includes generic syntax, so* every *relational theorem applies to clients that use $\Lambda/\cdot\,[\alpha_n]$ internally, which previously none did; and (ii) reuse at the typing level: one generic client value serves providers at different brands in one store (mechanized in the examples), which no concrete-signature client can.*

2 λ_{SCE} (SCE)

2.1 Syntax

$$
\begin{aligned}
T, A, B, R, S, \Gamma &::= \mathsf{Int} \mid \varepsilon \mid T \to T \mid T \,\&\, T \mid T \vee T \mid \{\ell : T\} \mid \mathsf{sig}\ \sigma \mid X_k \mid \mu T \mid \alpha_n \\
\sigma &::= \langle T \rangle \mid T \Rightarrow \sigma \\
s &::= \bullet \mid \circ \qquad \Delta \in \mathbb{N} \rightharpoonup \mathsf{Typ} \qquad \emptyset = \lambda n.\, \mathsf{none} \qquad \Delta \sqsubseteq \Delta' := \forall n\, R.\ \Delta\, n = R \Rightarrow \Delta'\, n = R \\
e &::= ? \mid e.k \mid i \mid \varepsilon \mid \lambda A.\, e \mid e_1 \triangleright e_2 \mid \langle e_1, \lambda A.\, e_2 \rangle \mid e_1\, e_2 \mid e_1 \# e_2 \mid \{\ell = e\} \mid e.\ell \\
&\mid \mathsf{struct}^s\, e \mid \mathsf{functor}^s(A).\, e \mid \langle e_1, \mathsf{functor}(A).\, e_2 \rangle \mid e_1 \ltimes e_2 \mid e_1 \bullet e_2 \mid e_1 \ltimes^* e_2 \\
&\mid e_1 \,,, e_2 \mid \mathsf{let}^A\, e_1\ \mathsf{in}\ e_2 \mid \mathsf{open}\ e_1\ \mathsf{in}\ e_2 \\
&\mid \mathsf{inl}^B\, e \mid \mathsf{inr}^A\, e \mid \mathsf{case}\ e\ \mathsf{of}\ e_1 \mid e_2 \mid \mu\lambda(A{\to}B).\, e \mid \langle e_1, \mu\lambda(A{\to}B).\, e_2 \rangle \mid \mathsf{fold}^T\, e \mid \mathsf{unfold}\ e \\
&\mid \langle n \rangle\, e \mid \mathsf{seal}\ n\ R\ S\ e \mid \mathsf{unseal}\ n\ R\ S\ e \\
v &::= i \mid \varepsilon \mid \langle v, \lambda A.e \rangle \mid \langle v, \mathsf{functor}(A).e \rangle \mid v_1 \# v_2 \mid \{\ell = v\} \\
&\mid \mathsf{inl}^B\, v \mid \mathsf{inr}^A\, v \mid \langle v, \mu\lambda(A{\to}B).e \rangle \mid \mathsf{fold}^T\, v \mid \langle n \rangle\, v
\end{aligned}
$$

$$
X_k[d{:=}S]^\mu = \begin{cases} S & k = d \\ X_k & k \neq d \end{cases} \qquad (\mu T)[d{:=}S]^\mu = \mu\,(T[d{+}1{:=}S]^\mu) \qquad \alpha_n[d{:=}S]^\mu = \alpha_n
$$

$$
\alpha_m[n{:=}R] = \begin{cases} R & m = n \\ \alpha_m & m \neq n \end{cases} \qquad X_k[n{:=}R] = X_k \qquad (\mu T)[n{:=}R] = \mu\,(T[n{:=}R])
$$

(both homomorphic otherwise, incl. σ)

$$
\frac{}{A\&B \ni_0 B}\ \textsc{zero} \qquad \frac{A \ni_k C}{A\&B \ni_{k+1} C}\ \textsc{succ} \qquad \frac{}{\ell \in \{\ell : T\}}\ \textsc{rcd} \qquad \frac{\ell \in A}{\ell \in A\&B}\ \textsc{andl} \qquad \frac{\ell \in B}{\ell \in A\&B}\ \textsc{andr}
$$

$$
\frac{\ell \in T}{\ell \in \mathsf{sig}\,\langle T \rangle}\ \textsc{sig} \qquad \frac{}{\ell : T \in \{\ell : T\}}\ \textsc{zero} \qquad \frac{\ell : T \in A \qquad \neg\, \ell \in B}{\ell : T \in A\&B}\ \textsc{andl} \qquad \frac{\ell : T \in B \qquad \neg\, \ell \in A}{\ell : T \in A\&B}\ \textsc{andr}
$$

$$
\frac{\ell : A \in T}{\ell : A \in \mathsf{sig}\,\langle T \rangle}\ \textsc{sig}
$$

2.2 Elaboration (SCE $\to \lambda_E^{\leq}$)

Type translation $\quad [\![\cdot]\!] : \mathsf{SCE.Typ} \to \mathsf{Seal.Typ}, \quad [\![\Delta]\!] := \lambda n.\ [\![\Delta\, n]\!]$

$$
\begin{aligned}
&[\![\mathsf{Int}]\!] = \mathsf{Int} \quad [\![\varepsilon]\!] = \varepsilon \quad [\![A \to B]\!] = [\![A]\!] \to [\![B]\!] \quad [\![A\&B]\!] = [\![A]\!] \,\&\, [\![B]\!] \quad [\![A \vee B]\!] = [\![A]\!] \vee [\![B]\!] \\
&[\![\{\ell : A\}]\!] = \{\ell : [\![A]\!]\} \quad [\![\mathsf{sig}\,\langle T \rangle]\!] = [\![T]\!] \quad [\![\mathsf{sig}\,(T \Rightarrow \sigma)]\!] = [\![T]\!] \to [\![\mathsf{sig}\,\sigma]\!] \\
&[\![X_k]\!] = X_k \quad [\![\mu T]\!] = \mu\,[\![T]\!] \quad [\![\alpha_n]\!] = \alpha_n
\end{aligned}
$$

Import packages $\quad \mathsf{wire}(\Gamma, c_1, D), \quad \mathsf{WireOk}\ \Gamma\ \Gamma_1\ D$

$$
\begin{aligned}
&\mathsf{wire}(\Gamma, c_1, \{\ell : A\}) = \{\ell = c_1.\ell\} \qquad \mathsf{wire}(\Gamma, c_1, _) = \varepsilon \\
&\mathsf{wire}(\Gamma, c_1, D\&\{\ell : A\}) = \mathsf{wire}(\Gamma, c_1, D) \,\#\, \big((? : \Gamma) \triangleright \{\ell = c_1.\ell\}\big)
\end{aligned}
$$

$$
\frac{\ell : A \in \Gamma_1}{\mathsf{WireOk}\ \Gamma\ \Gamma_1\ \{\ell : A\}}\ \textsc{one} \qquad \frac{\mathsf{WireOk}\ \Gamma\ \Gamma_1\ D \qquad \ell : A \in \Gamma_1 \qquad [\![D]\!] * \Gamma \qquad [\![D]\!] * \{\ell : [\![A]\!]\}}{\mathsf{WireOk}\ \Gamma\ \Gamma_1\ (D\&\{\ell : A\})}\ \textsc{more}
$$

Elaboration $\Delta; \Gamma \vdash e : A \rightsquigarrow \overline{e}$

$$\frac{}{\Delta; \Gamma \vdash ? : \Gamma \rightsquigarrow ?}\ \textsc{equery} \qquad \frac{}{\Delta; \Gamma \vdash i : \mathsf{Int} \rightsquigarrow i}\ \textsc{elit} \qquad \frac{}{\Delta; \Gamma \vdash \varepsilon : \varepsilon \rightsquigarrow \varepsilon}\ \textsc{eunit}$$

$$\frac{\Delta; \Gamma \vdash e_1 : A \to B \rightsquigarrow \overline{e}_1 \qquad \Delta; \Gamma \vdash e_2 : A \rightsquigarrow \overline{e}_2}{\Delta; \Gamma \vdash e_1\, e_2 : B \rightsquigarrow \overline{e}_1\, \overline{e}_2}\ \textsc{eapp} \qquad \frac{\Delta; \Gamma \vdash e : A \rightsquigarrow \overline{e} \qquad A \ni_k B}{\Delta; \Gamma \vdash e.k : B \rightsquigarrow \overline{e}.k}\ \textsc{eproj}$$

$$\frac{\Delta; \Gamma \vdash e_1 : \Gamma' \rightsquigarrow \overline{e}_1 \qquad \Delta; \Gamma' \vdash e_2 : A \rightsquigarrow \overline{e}_2}{\Delta; \Gamma \vdash e_1 \triangleright e_2 : A \rightsquigarrow \overline{e}_1 \triangleright \overline{e}_2}\ \textsc{ebox}$$

$$\frac{\Delta; \Gamma \vdash e_1 : A \rightsquigarrow \overline{e}_1 \qquad \Delta; \Gamma \& A \vdash e_2 : B \rightsquigarrow \overline{e}_2 \qquad [\![A]\!] * [\![\Gamma]\!] \qquad [\![A]\!] * [\![B]\!]}{\Delta; \Gamma \vdash e_1 \# e_2 : A \& B \rightsquigarrow \overline{e}_1 \# \overline{e}_2}\ \textsc{edmrg}$$

$$\frac{v_1\ \mathsf{value} \qquad v_2\ \mathsf{value} \qquad \Delta; \varepsilon \vdash v_1 : A \rightsquigarrow \overline{w}_1 \qquad \Delta; \varepsilon \vdash v_2 : B \rightsquigarrow \overline{w}_2 \qquad [\![A]\!] * [\![B]\!]}{\Delta; \Gamma \vdash v_1 \# v_2 : A \& B \rightsquigarrow \overline{w}_1 \# \overline{w}_2}\ \textsc{evmrg}$$

$$\frac{\Delta; \Gamma \vdash e_1 : A \rightsquigarrow \overline{e}_1 \qquad \Delta; \Gamma \vdash e_2 : B \rightsquigarrow \overline{e}_2 \qquad [\![A]\!] * [\![\Gamma]\!] \qquad [\![A]\!] * [\![B]\!]}{\Delta; \Gamma \vdash e_1 \,,, e_2 : A \& B \rightsquigarrow \overline{e}_1 \# \left((? : [\![\Gamma]\!]) \triangleright \overline{e}_2\right)}\ \textsc{enmrg}$$

$$\frac{\Delta; \Gamma \& A \vdash e : B \rightsquigarrow \overline{e} \qquad [\![\Gamma]\!] * [\![A]\!]}{\Delta; \Gamma \vdash \lambda A.e : A \to B \rightsquigarrow \lambda([\![A]\!] \to [\![B]\!]).\overline{e}}\ \textsc{elam} \qquad \frac{\Delta; \Gamma \vdash e : B \rightsquigarrow \overline{e} \qquad \ell : A \in B}{\Delta; \Gamma \vdash e.\ell : A \rightsquigarrow \overline{e}.\ell}\ \textsc{erproj}$$

$$\frac{v_1\ \mathsf{value} \qquad \Delta; \varepsilon \vdash v_1 : \Gamma' \rightsquigarrow \overline{v}_1 \qquad \Delta; \Gamma' \& A \vdash e_2 : B \rightsquigarrow \overline{e}_2 \qquad [\![\Gamma']\!] * [\![A]\!]}{\Delta; \Gamma \vdash \langle v_1, \lambda A.e_2 \rangle : A \to B \rightsquigarrow \langle \overline{v}_1, \lambda([\![A]\!] \to [\![B]\!]).\overline{e}_2 \rangle}\ \textsc{eclos}$$

$$\frac{\Delta; \Gamma \vdash e : A \rightsquigarrow \overline{e}}{\Delta; \Gamma \vdash \{\ell = e\} : \{\ell : A\} \rightsquigarrow \{\ell = \overline{e}\}}\ \textsc{elrec} \qquad \frac{\Delta; \Gamma \vdash e_1 : A \rightsquigarrow \overline{e}_1 \qquad \Delta; \Gamma \& A \vdash e_2 : B \rightsquigarrow \overline{e}_2 \qquad [\![\Gamma]\!] * [\![A]\!]}{\Delta; \Gamma \vdash \mathsf{let}^A e_1\ \mathsf{in}\ e_2 : B \rightsquigarrow \left(\lambda([\![A]\!] \to [\![B]\!]).\overline{e}_2\right) \overline{e}_1}\ \textsc{eletb}$$

$$\frac{\Delta; \Gamma \vdash e_1 : \{\ell : A\} \rightsquigarrow \overline{e}_1 \qquad \Delta; \Gamma \& A \vdash e_2 : B \rightsquigarrow \overline{e}_2 \qquad [\![\Gamma]\!] * [\![A]\!]}{\Delta; \Gamma \vdash \mathsf{open}\ e_1\ \mathsf{in}\ e_2 : B \rightsquigarrow \left(\lambda([\![A]\!] \to [\![B]\!]).\overline{e}_2\right) (\overline{e}_1.\ell)}\ \textsc{eopenm}$$

$$\frac{\Delta; \Gamma \vdash e : A \rightsquigarrow \overline{e}}{\Delta; \Gamma \vdash \mathsf{inl}^B e : A \vee B \rightsquigarrow \mathsf{inl}^{[\![B]\!]}\, \overline{e}}\ \textsc{einl} \qquad \frac{\Delta; \Gamma \vdash e : B \rightsquigarrow \overline{e}}{\Delta; \Gamma \vdash \mathsf{inr}^A e : A \vee B \rightsquigarrow \mathsf{inr}^{[\![A]\!]}\, \overline{e}}\ \textsc{einr}$$

$$\frac{\Delta; \Gamma \vdash e : A \vee B \rightsquigarrow \overline{e} \qquad \Delta; \Gamma \& A \vdash e_1 : C \rightsquigarrow \overline{e}_1 \qquad \Delta; \Gamma \& B \vdash e_2 : C \rightsquigarrow \overline{e}_2 \qquad [\![\Gamma]\!] * [\![A]\!] \qquad [\![\Gamma]\!] * [\![B]\!]}{\Delta; \Gamma \vdash \mathsf{case}\ e\ \mathsf{of}\ e_1 \mid e_2 : C \rightsquigarrow \mathsf{case}\ \overline{e}\ \mathsf{of}\ \overline{e}_1 \mid \overline{e}_2}\ \textsc{ecase}$$

$$\frac{\Delta; (\Gamma \& (A \to B)) \& A \vdash e : B \rightsquigarrow \overline{e} \qquad [\![\Gamma]\!] * ([\![A]\!] \to [\![B]\!]) \qquad ([\![\Gamma]\!] \& ([\![A]\!] \to [\![B]\!])) * [\![A]\!]}{\Delta; \Gamma \vdash \mu\lambda(A \to B).e : A \to B \rightsquigarrow \mu\lambda([\![A]\!] \to [\![B]\!]).\overline{e}}\ \textsc{eflam}$$

$$\frac{\begin{array}{c} v_1\ \mathsf{value} \qquad \Delta; \varepsilon \vdash v_1 : \Gamma' \rightsquigarrow \overline{v}_1 \\ \Delta; (\Gamma' \& (A \to B)) \& A \vdash e_2 : B \rightsquigarrow \overline{e}_2 \qquad [\![\Gamma']\!] * ([\![A]\!] \to [\![B]\!]) \qquad ([\![\Gamma']\!] \& ([\![A]\!] \to [\![B]\!])) * [\![A]\!] \end{array}}{\Delta; \Gamma \vdash \langle v_1, \mu\lambda(A \to B).e_2 \rangle : A \to B \rightsquigarrow \langle \overline{v}_1, \mu\lambda([\![A]\!] \to [\![B]\!]).\, \overline{e}_2 \rangle^{[\![B]\!]}}\ \textsc{efclos}$$

$$\frac{\Delta; \Gamma \vdash e : T[X_0 := \mu T]^\mu \rightsquigarrow \overline{e}}{\Delta; \Gamma \vdash \mathsf{fold}^T e : \mu T \rightsquigarrow \mathsf{fold}^{[\![T]\!]}\, \overline{e}}\ \textsc{efold} \qquad \frac{\Delta; \Gamma \vdash e : \mu T \rightsquigarrow \overline{e} \qquad A = T[X_0 := \mu T]^\mu}{\Delta; \Gamma \vdash \mathsf{unfold}\ e : A \rightsquigarrow \mathsf{unfold}\ \overline{e}}\ \textsc{eunfold}$$

$$\frac{\Delta;\varepsilon \vdash e : B \rightsquigarrow \overline{e}}{\Delta;\Gamma \vdash \mathsf{struct}^{\bullet}\ e : B \rightsquigarrow \varepsilon \triangleright \overline{e}}\ \textsc{emstruct-}\bullet \qquad \frac{\Delta;\Gamma \vdash e : B \rightsquigarrow \overline{e}}{\Delta;\Gamma \vdash \mathsf{struct}^{\circ}\ e : B \rightsquigarrow ? \triangleright \overline{e}}\ \textsc{emstruct-}\circ$$

$$\frac{\Delta;\varepsilon \& A \vdash e : B \rightsquigarrow \overline{e}}{\Delta;\Gamma \vdash \mathsf{functor}^{\bullet}(A).e : \mathsf{sig}\,(A \Rightarrow \langle B\rangle) \rightsquigarrow \varepsilon \triangleright \lambda([\![A]\!]{\to}[\![B]\!]).\overline{e}}\ \textsc{emfunctor-}\bullet$$

$$\frac{\Delta;\Gamma \& A \vdash e : B \rightsquigarrow \overline{e} \qquad [\![\Gamma]\!] * [\![A]\!]}{\Delta;\Gamma \vdash \mathsf{functor}^{\circ}(A).e : \mathsf{sig}\,(A \Rightarrow \langle B\rangle) \rightsquigarrow \lambda([\![A]\!]{\to}[\![B]\!]).\overline{e}}\ \textsc{emfunctor-}\circ$$

$$\frac{v_1\ \mathsf{value} \qquad \Delta;\varepsilon \vdash v_1 : \Gamma' \rightsquigarrow \overline{v}_1 \qquad \Delta;\Gamma' \& A \vdash e_2 : B \rightsquigarrow \overline{e}_2 \qquad [\![\Gamma']\!] * [\![A]\!]}{\Delta;\Gamma \vdash \langle v_1, \mathsf{functor}(A).e_2\rangle : \mathsf{sig}\,(A \Rightarrow \langle B\rangle) \rightsquigarrow \langle \overline{v}_1, \lambda([\![A]\!]{\to}[\![B]\!]).\overline{e}_2\rangle}\ \textsc{emclos}$$

$$\frac{\Delta;\Gamma \vdash e_1 : \mathsf{sig}\,(A \Rightarrow \langle B\rangle) \rightsquigarrow \overline{e}_1 \qquad \Delta;\Gamma \vdash e_2 : A \rightsquigarrow \overline{e}_2}{\Delta;\Gamma \vdash e_1 \bullet e_2 : B \rightsquigarrow \overline{e}_1\,\overline{e}_2}\ \textsc{emapp}$$

$$\frac{\Delta;\Gamma \vdash e_1 : \Gamma_1 \rightsquigarrow \overline{e}_1 \qquad \Delta;\Gamma \vdash e_2 : \mathsf{sig}\,(\{\ell : A\} \Rightarrow \langle B\rangle) \rightsquigarrow \overline{e}_2 \qquad \ell : A \in \Gamma_1 \qquad [\![\Gamma_1]\!] * [\![\Gamma]\!] \qquad [\![\Gamma_1]\!] * [\![B]\!]}{\Delta;\Gamma \vdash e_1 \ltimes e_2 : \Gamma_1 \& B \rightsquigarrow \overline{e}_1 \,\#\, \Big((? : [\![\Gamma]\!]) \triangleright \big(\overline{e}_2\ \{\ell = \overline{e}_1.\ell\}\big)\Big)}\ \textsc{emlink}$$

$$\frac{\begin{array}{c}\Delta;\Gamma \vdash e_1 : \Gamma_1 \rightsquigarrow \overline{e}_1 \\ \Delta;\Gamma \vdash e_2 : \mathsf{sig}\,(D \Rightarrow \langle B\rangle) \rightsquigarrow \overline{e}_2 \qquad \mathsf{WireOk}\ [\![\Gamma]\!]\ \Gamma_1\ D \qquad [\![\Gamma_1]\!] * [\![\Gamma]\!] \qquad [\![\Gamma_1]\!] * [\![B]\!]\end{array}}{\Delta;\Gamma \vdash e_1 \ltimes^{*} e_2 : \Gamma_1 \& B \rightsquigarrow \overline{e}_1 \,\#\, \Big((? : [\![\Gamma]\!]) \triangleright \big(\overline{e}_2\ \mathsf{wire}([\![\Gamma]\!], \overline{e}_1, D)\big)\Big)}\ \textsc{emlinkn}$$

$$\frac{\Delta\, n = R \qquad v\ \mathsf{value} \qquad \Delta;\varepsilon \vdash v : R \rightsquigarrow \overline{w}}{\Delta;\Gamma \vdash \langle n\rangle\, v : \alpha_n \rightsquigarrow \langle n\rangle\, \overline{w}}\ \textsc{ewrap}$$

$$\frac{\Delta\, n = R \qquad \mathsf{NoRes}\ [\![R]\!] \qquad \mathsf{WfSig}\ n\ [\![R]\!]\ [\![S]\!] \qquad \Delta;\Gamma \vdash e : S[n{:=}R] \rightsquigarrow \overline{e}}{\Delta;\Gamma \vdash \mathsf{seal}\ n\ R\ S\ e : S \rightsquigarrow \mathsf{seal}\ n\ [\![R]\!]\ [\![S]\!]\ \overline{e}}\ \textsc{emseal}$$

$$\frac{\Delta\, n = R \qquad \mathsf{NoRes}\ [\![R]\!] \qquad \mathsf{WfSig}\ n\ [\![R]\!]\ [\![S]\!] \qquad \Delta;\Gamma \vdash e : S \rightsquigarrow \overline{e}}{\Delta;\Gamma \vdash \mathsf{unseal}\ n\ R\ S\ e : S[n{:=}R] \rightsquigarrow \mathsf{unseal}\ n\ [\![R]\!]\ [\![S]\!]\ \overline{e}}\ \textsc{emunseal}$$

2.3 Semantics

Value lookups, selection, packages $\quad v \ni_k v', \quad v.\ell \rightsquigarrow v', \quad v \gg_D p$

$$\frac{}{v_1 \,\#\, v_2 \ni_0 v_2}\ \textsc{dmrg-zero} \qquad \frac{v_1 \ni_k v'}{v_1 \,\#\, v_2 \ni_{k+1} v'}\ \textsc{dmrg-succ} \qquad \frac{}{v_1 \,,,\, v_2 \ni_0 v_2}\ \textsc{nmrg-zero}$$

$$\frac{v_1 \ni_k v'}{v_1 \,,,\, v_2 \ni_{k+1} v'}\ \textsc{nmrg-succ} \qquad \frac{}{\{\ell = v\}.\ell \rightsquigarrow v}\ \textsc{rcd} \qquad \frac{v_1.\ell \rightsquigarrow v'}{v_1 \,\#\, v_2.\ell \rightsquigarrow v'}\ \textsc{dmrg-left}$$

$$\frac{v_2.\ell \rightsquigarrow v'}{v_1 \,\#\, v_2.\ell \rightsquigarrow v'}\ \textsc{dmrg-right} \qquad \frac{v_1.\ell \rightsquigarrow v'}{v_1 \,,,\, v_2.\ell \rightsquigarrow v'}\ \textsc{nmrg-left} \qquad \frac{v_2.\ell \rightsquigarrow v'}{v_1 \,,,\, v_2.\ell \rightsquigarrow v'}\ \textsc{nmrg-right}$$

$$\frac{v.\ell \rightsquigarrow v_\ell}{v \gg_{\{\ell:A\}} \{\ell = v_\ell\}}\ \textsc{one} \qquad \frac{v \gg_D p \qquad v.\ell \rightsquigarrow v_\ell}{v \gg_{D\&\{\ell:A\}} p \,\#\, \{\ell = v_\ell\}}\ \textsc{more}$$

Sealing coercions $v \Downarrow_S^{\mathsf{seal}\,n\,R} w$, $v \Downarrow_S^{\mathsf{unseal}\,n\,R} w$ ($\mathsf{pxe}(c) := \varepsilon \# \{\#\mathtt{f} = c\}$, $\mathsf{pxf} := (?.1).\#\mathtt{f}$)

$$\frac{}{v \Downarrow_{\alpha_n}^{\mathsf{seal}\,n\,R} \langle n \rangle\, v}\ \text{BRAND-EQ} \qquad \frac{m \neq n}{v \Downarrow_{\alpha_m}^{\mathsf{seal}\,n\,R} v}\ \text{BRAND-NE} \qquad \frac{}{i \Downarrow_{\mathsf{Int}}^{\mathsf{seal}\,n\,R} i}\ \text{INT} \qquad \frac{}{v \Downarrow_{\varepsilon}^{\mathsf{seal}\,n\,R} \varepsilon}\ \text{TOP}$$

$$\frac{v_1 \Downarrow_A^{\mathsf{seal}\,n\,R} w_1 \qquad v_2 \Downarrow_B^{\mathsf{seal}\,n\,R} w_2}{v_1 \# v_2 \Downarrow_{A\&B}^{\mathsf{seal}\,n\,R} w_1 \# w_2}\ \text{AND} \qquad \frac{v \Downarrow_A^{\mathsf{seal}\,n\,R} w}{\{\ell = v\} \Downarrow_{\{\ell:A\}}^{\mathsf{seal}\,n\,R} \{\ell = w\}}\ \text{RCD}$$

$$\frac{}{c \Downarrow_{A\to B}^{\mathsf{seal}\,n\,R} \langle \mathsf{pxe}(c), \lambda A.\ \mathsf{seal}\ n\ R\ B\ (\mathsf{pxf}\ (\mathsf{unseal}\ n\ R\ A\ ?.0))\rangle}\ \text{ARR}$$

$$\frac{}{\langle n \rangle\, v \Downarrow_{\alpha_n}^{\mathsf{unseal}\,n\,R} v}\ \text{BRAND-EQ} \qquad \frac{m \neq n}{v \Downarrow_{\alpha_m}^{\mathsf{unseal}\,n\,R} v}\ \text{BRAND-NE} \qquad \frac{}{i \Downarrow_{\mathsf{Int}}^{\mathsf{unseal}\,n\,R} i}\ \text{INT} \qquad \frac{}{v \Downarrow_{\varepsilon}^{\mathsf{unseal}\,n\,R} \varepsilon}\ \text{TOP}$$

$$\frac{v_1 \Downarrow_A^{\mathsf{unseal}\,n\,R} w_1 \qquad v_2 \Downarrow_B^{\mathsf{unseal}\,n\,R} w_2}{v_1 \# v_2 \Downarrow_{A\&B}^{\mathsf{unseal}\,n\,R} w_1 \# w_2}\ \text{AND} \qquad \frac{v \Downarrow_A^{\mathsf{unseal}\,n\,R} w}{\{\ell = v\} \Downarrow_{\{\ell:A\}}^{\mathsf{unseal}\,n\,R} \{\ell = w\}}\ \text{RCD}$$

$$\frac{}{c \Downarrow_{A\to B}^{\mathsf{unseal}\,n\,R} \langle \mathsf{pxe}(c), \lambda A[n{:=}R].\ \mathsf{unseal}\ n\ R\ B\ (\mathsf{pxf}\ (\mathsf{seal}\ n\ R\ A\ ?.0))\rangle}\ \text{ARR}$$

Big-step evaluation $\rho \vdash e \Downarrow v$ **(all rules carry ρ value; value premises shown)**

$$\frac{}{\rho \vdash ? \Downarrow \rho}\ \textsc{query} \qquad \frac{}{\rho \vdash i \Downarrow i}\ \textsc{lit} \qquad \frac{}{\rho \vdash \varepsilon \Downarrow \varepsilon}\ \textsc{unit} \qquad \frac{v\ \mathsf{value}}{\rho \vdash \langle v, \lambda A.e\rangle \Downarrow \langle v, \lambda A.e\rangle}\ \textsc{clos-val}$$

$$\frac{v\ \mathsf{value}}{\rho \vdash \langle v, \mathsf{functor}(A).e\rangle \Downarrow \langle v, \mathsf{functor}(A).e\rangle}\ \textsc{mclos-val} \qquad \frac{\rho \vdash e \Downarrow v \qquad v \ni_k v'}{\rho \vdash e.k \Downarrow v'}\ \textsc{proj} \qquad \frac{}{\rho \vdash \lambda A.e \Downarrow \langle \rho, \lambda A.e\rangle}\ \textsc{lam}$$

$$\frac{\rho \vdash e_1 \Downarrow v_1 \qquad v_1 \vdash e_2 \Downarrow v}{\rho \vdash e_1 \triangleright e_2 \Downarrow v}\ \textsc{box} \qquad \frac{\rho \vdash e_1 \Downarrow \langle v_1, \lambda A.e\rangle \qquad \rho \vdash e_2 \Downarrow v_2 \qquad v_1 \# v_2 \vdash e \Downarrow v}{\rho \vdash e_1\, e_2 \Downarrow v}\ \textsc{app-clos}$$

$$\frac{\rho \vdash e_1 \Downarrow \langle v_1, \mu\lambda(A{\to}B).e\rangle \qquad \rho \vdash e_2 \Downarrow v_2 \qquad (v_1 \# \langle v_1, \mu\lambda(A{\to}B).e\rangle) \# v_2 \vdash e \Downarrow v}{\rho \vdash e_1\, e_2 \Downarrow v}\ \textsc{app-fclos}$$

$$\frac{\rho \vdash e_1 \Downarrow \langle v_1, \mathsf{functor}(A).e\rangle \qquad \rho \vdash e_2 \Downarrow v_2 \qquad v_1 \# v_2 \vdash e \Downarrow v}{\rho \vdash e_1 \bullet e_2 \Downarrow v}\ \textsc{app-mclos}$$

$$\frac{\rho \vdash e_1 \Downarrow v_1 \qquad \rho \# v_1 \vdash e_2 \Downarrow v_2}{\rho \vdash e_1 \# e_2 \Downarrow v_1 \# v_2}\ \textsc{dmrg} \qquad \frac{\rho \vdash e_1 \Downarrow v_1 \qquad \rho \vdash e_2 \Downarrow v_2}{\rho \vdash e_1 \,,, e_2 \Downarrow v_1 \# v_2}\ \textsc{nmrg} \qquad \frac{\rho \vdash e \Downarrow v}{\rho \vdash \{\ell = e\} \Downarrow \{\ell = v\}}\ \textsc{lrec}$$

$$\frac{\rho \vdash e \Downarrow v \qquad v.\ell \rightsquigarrow v'}{\rho \vdash e.\ell \Downarrow v'}\ \textsc{rproj} \qquad \frac{\rho \vdash e_1 \Downarrow v_1 \qquad \rho \# v_1 \vdash e_2 \Downarrow v}{\rho \vdash \mathsf{let}^A e_1\ \mathsf{in}\ e_2 \Downarrow v}\ \textsc{letb}$$

$$\frac{\rho \vdash e_1 \Downarrow \{\ell = v'\} \qquad \rho \# v' \vdash e_2 \Downarrow v}{\rho \vdash \mathsf{open}\ e_1\ \mathsf{in}\ e_2 \Downarrow v}\ \textsc{openm} \qquad \frac{\varepsilon \vdash e \Downarrow v}{\rho \vdash \mathsf{struct}^\bullet\, e \Downarrow v}\ \textsc{mstruct-}\bullet \qquad \frac{\rho \vdash e \Downarrow v}{\rho \vdash \mathsf{struct}^\circ\, e \Downarrow v}\ \textsc{mstruct-}\circ$$

$$\frac{}{\rho \vdash \mathsf{functor}^\bullet(A).e \Downarrow \langle \varepsilon, \mathsf{functor}(A).e\rangle}\ \textsc{mfunctor-}\bullet \qquad \frac{}{\rho \vdash \mathsf{functor}^\circ(A).e \Downarrow \langle \rho, \mathsf{functor}(A).e\rangle}\ \textsc{mfunctor-}\circ$$

$$\frac{\rho \vdash e_1 \Downarrow v_1 \qquad \rho \vdash e_2 \Downarrow \langle v_2, \mathsf{functor}(\{\ell : A\}).e\rangle \qquad v_1.\ell \rightsquigarrow v_\ell \qquad v_2 \# \{\ell = v_\ell\} \vdash e \Downarrow v_3}{\rho \vdash e_1 \ltimes e_2 \Downarrow v_1 \# v_3}\ \textsc{mlink}$$

$$\frac{\rho \vdash e_1 \Downarrow v_1 \qquad \rho \vdash e_2 \Downarrow \langle v_2, \mathsf{functor}(D).e\rangle \qquad v_1 \gg_D p \qquad v_2 \# p \vdash e \Downarrow v_3}{\rho \vdash e_1 \ltimes^* e_2 \Downarrow v_1 \# v_3}\ \textsc{mlinkn} \qquad \frac{\rho \vdash e \Downarrow v}{\rho \vdash \mathsf{inl}^B e \Downarrow \mathsf{inl}^B v}\ \textsc{inl}$$

$$\frac{\rho \vdash e \Downarrow v}{\rho \vdash \mathsf{inr}^A e \Downarrow \mathsf{inr}^A v}\ \textsc{inr} \qquad \frac{\rho \vdash e \Downarrow \mathsf{inl}^B v_1 \qquad \rho \# v_1 \vdash e_1 \Downarrow v}{\rho \vdash \mathsf{case}\ e\ \mathsf{of}\ e_1 \mid e_2 \Downarrow v}\ \textsc{case-inl}$$

$$\frac{\rho \vdash e \Downarrow \mathsf{inr}^A v_1 \qquad \rho \# v_1 \vdash e_2 \Downarrow v}{\rho \vdash \mathsf{case}\ e\ \mathsf{of}\ e_1 \mid e_2 \Downarrow v}\ \textsc{case-inr} \qquad \frac{v\ \mathsf{value}}{\rho \vdash \langle v, \mu\lambda(A{\to}B).e\rangle \Downarrow \langle v, \mu\lambda(A{\to}B).e\rangle}\ \textsc{fclos-val}$$

$$\frac{}{\rho \vdash \mu\lambda(A{\to}B).e \Downarrow \langle \rho, \mu\lambda(A{\to}B).e\rangle}\ \textsc{flam} \qquad \frac{\rho \vdash e \Downarrow v}{\rho \vdash \mathsf{fold}^T e \Downarrow \mathsf{fold}^T v}\ \textsc{fold} \qquad \frac{\rho \vdash e \Downarrow \mathsf{fold}^T v}{\rho \vdash \mathsf{unfold}\ e \Downarrow v}\ \textsc{unfold}$$

$$\frac{\rho \vdash e \Downarrow v}{\rho \vdash \langle n\rangle\, e \Downarrow \langle n\rangle\, v}\ \textsc{wrap} \qquad \frac{\rho \vdash e \Downarrow v \qquad v \Downarrow_S^{\mathsf{seal}\, n\, R} w}{\rho \vdash \mathsf{seal}\ n\ R\ S\ e \Downarrow w}\ \textsc{mseal} \qquad \frac{\rho \vdash e \Downarrow v \qquad v \Downarrow_S^{\mathsf{unseal}\, n\, R} w}{\rho \vdash \mathsf{unseal}\ n\ R\ S\ e \Downarrow w}\ \textsc{munseal}$$

2.4 Meta-theory

Definition 2.1 (EVal $\Delta \vdash v \sim_A \overline{v}$).

$$\frac{}{\Delta \vdash i \sim_{\mathsf{Int}} i}\ \text{LIT} \qquad \frac{}{\Delta \vdash \varepsilon \sim_{\varepsilon} \varepsilon}\ \text{UNIT} \qquad \frac{\Delta \vdash v_1 \sim_A \overline{w}_1 \qquad \Delta \vdash v_2 \sim_B \overline{w}_2 \qquad [\![A]\!] * [\![B]\!]}{\Delta \vdash v_1 \# v_2 \sim_{A\&B} \overline{w}_1 \# \overline{w}_2}\ \text{MRG}$$

$$\frac{\Delta \vdash v \sim_A \overline{w}}{\Delta \vdash \{\ell = v\} \sim_{\{\ell:A\}} \{\ell = \overline{w}\}}\ \text{RCD} \qquad \frac{\rho\ \mathsf{value} \qquad \Delta \vdash \rho \sim_{\Gamma'} \overline{\rho} \qquad \Delta; \Gamma' \& A \vdash e : B \rightsquigarrow \overline{e} \qquad [\![\Gamma']\!] * [\![A]\!]}{\Delta \vdash \langle \rho, \lambda A.e \rangle \sim_{A \to B} \langle \overline{\rho}, \lambda([\![A]\!] \to [\![B]\!]).\overline{e} \rangle}\ \text{CLOS}$$

$$\frac{\rho\ \mathsf{value} \qquad \Delta \vdash \rho \sim_{\Gamma'} \overline{\rho} \qquad \Delta; \Gamma' \& A \vdash e : B \rightsquigarrow \overline{e} \qquad [\![\Gamma']\!] * [\![A]\!]}{\Delta \vdash \langle \rho, \mathsf{functor}(A).e \rangle \sim_{\mathsf{sig}\,(A \Rightarrow \langle B \rangle)} \langle \overline{\rho}, \lambda([\![A]\!] \to [\![B]\!]).\overline{e} \rangle}\ \text{MCLOS}$$

$$\frac{\lceil [\![A]\!] \rceil \qquad v\ \mathsf{value} \qquad \Delta; \varepsilon \vdash v : A \rightsquigarrow \overline{w}}{\Delta \vdash v \sim_A [\![A]\!]^{\uparrow}}\ \text{GEN} \qquad \frac{\Delta\, n = R \qquad \Delta \vdash v \sim_R \overline{w}}{\Delta \vdash \langle n \rangle\, v \sim_{\alpha_n} \langle n \rangle\, \overline{w}}\ \text{WRAP}$$

$$\frac{\Delta \vdash v \sim_A \overline{w}}{\Delta \vdash \mathsf{inl}^B v \sim_{A \vee B} \mathsf{inl}^{[\![B]\!]}\, \overline{w}}\ \text{INL} \qquad \frac{\Delta \vdash v \sim_B \overline{w}}{\Delta \vdash \mathsf{inr}^A v \sim_{A \vee B} \mathsf{inr}^{[\![A]\!]}\, \overline{w}}\ \text{INR}$$

$$\frac{\Delta \vdash \rho \sim_{\Gamma'} \overline{\rho} \qquad \Delta; (\Gamma' \& (A \to B)) \& A \vdash e : B \rightsquigarrow \overline{e} \qquad [\![\Gamma']\!] * ([\![A]\!] \to [\![B]\!]) \qquad ([\![\Gamma']\!] \& ([\![A]\!] \to [\![B]\!])) * [\![A]\!] \qquad \rho\ \mathsf{value}}{\Delta \vdash \langle \rho, \mu\lambda(A \to B).e \rangle \sim_{A \to B} \langle \overline{\rho}, \mu\lambda([\![A]\!] \to [\![B]\!]).\, \overline{e} \rangle^{[\![B]\!]}}\ \text{FCLOS}$$

$$\frac{\Delta \vdash v \sim_{T[X_0 := \mu T]^{\mu}} \overline{w}}{\Delta \vdash \mathsf{fold}^T v \sim_{\mu T} \mathsf{fold}^{[\![T]\!]}\, \overline{w}}\ \text{FOLD}$$

Theorem 2.2 (seal_type_preservation).

$$\Delta; \Gamma \vdash e : A \rightsquigarrow \overline{e} \implies [\![\Delta]\!]; [\![\Gamma]\!] \vdash \overline{e} : [\![A]\!]$$

Theorem 2.3 (elabSeal_weaken_store).

$$\Delta \sqsubseteq \Delta' \ \wedge\ \Delta; \Gamma \vdash e : A \rightsquigarrow \overline{e} \implies \Delta'; \Gamma \vdash e : A \rightsquigarrow \overline{e}$$

Theorem 2.4 (elabSeal_uniqueness). *Elaboration is a function of the term and (for non-values) its context:*

$$\Delta; \Gamma_1 \vdash e : T_1 \rightsquigarrow \overline{e}_1 \ \wedge\ \Delta; \Gamma_2 \vdash e : T_2 \rightsquigarrow \overline{e}_2 \ \wedge\ (\Gamma_1 = \Gamma_2 \vee e\ \mathsf{value}) \implies T_1 = T_2 \wedge \overline{e}_1 = \overline{e}_2$$

Theorem 2.5 (source_sprogress, source_spreservation, source_bigstep_deterministic). *Pure source metatheory, with elaborability playing the role of typing (SStep is the source small-step, BStep the big-step):*

$$(\exists \overline{e}.\ \Delta; \varepsilon \vdash e : A \rightsquigarrow \overline{e}) \implies e\ \mathsf{value} \vee \exists e'.\ \mathsf{SStep}\ \varepsilon\ e\ e'$$
$$(\exists \overline{e}.\ \Delta; \varepsilon \vdash e : A \rightsquigarrow \overline{e}) \ \wedge\ \mathsf{SStep}\ \varepsilon\ e\ e' \implies \exists \overline{e}'.\ \Delta; \varepsilon \vdash e' : A \rightsquigarrow \overline{e}'$$
$$(\exists \overline{e}.\ \Delta; \varepsilon \vdash e : A \rightsquigarrow \overline{e}) \ \wedge\ \varepsilon \vdash e \Downarrow v_1 \ \wedge\ \varepsilon \vdash e \Downarrow v_2 \implies v_1 = v_2$$

(Each also holds generalized over a well-typed environment; sealed-unit runs are covered — the sealing cases of preservation and progress go through the source coercions.)

Theorem 2.6 (seal_semantic_preservation).

$$\rho \vdash e \Downarrow v \ \wedge\ \Delta; \Gamma \vdash e : A \rightsquigarrow \overline{e} \ \wedge\ \rho\ \mathsf{value} \ \wedge\ \Delta \vdash \rho \sim_{\Gamma} \overline{\rho} \implies \exists \overline{v}.\ \overline{\rho} \vdash \overline{e} \hookrightarrow^* \overline{v} \ \wedge\ \Delta \vdash v \sim_A \overline{v}$$

Theorem 2.7 (seal_whole_program_correctness).

$$\Delta; \varepsilon \vdash e : A \rightsquigarrow \overline{e} \ \wedge\ \varepsilon \vdash e \Downarrow v \implies \exists \overline{v}.\ \varepsilon \vdash \overline{e} \hookrightarrow^* \overline{v} \ \wedge\ \Delta \vdash v \sim_A \overline{v}$$

Theorem 2.8 (seal_separate_compilation).

$$\Delta;\Gamma \vdash e_1 : \Gamma_1 \rightsquigarrow \overline{e}_1 \ \wedge\ \Delta;\Gamma \vdash e_2 : \mathsf{sig}\,(\{\ell : A\} \Rightarrow \langle B\rangle) \rightsquigarrow \overline{e}_2 \ \wedge\ \ell : A \in \Gamma_1 \ \wedge\ [\![\Gamma_1]\!] * [\![\Gamma]\!] \ \wedge\ [\![\Gamma_1]\!] * [\![B]\!] \wedge$$
$$\rho \vdash e_1 \ltimes e_2 \Downarrow v \ \wedge\ \rho\ \mathsf{value} \ \wedge\ \Delta \vdash \rho \sim_\Gamma \overline{\rho}$$
$$\Longrightarrow\ \exists \overline{v}.\ \overline{\rho} \vdash \overline{e}_1 \,\#\, \big((? : [\![\Gamma]\!]) \triangleright (\overline{e}_2\ \{\ell = \overline{e}_1.\ell\})\big) \hookrightarrow^* \overline{v} \ \wedge\ \Delta \vdash v \sim_{\Gamma_1 \& B} \overline{v}$$

Theorem 2.9 (seal_separate_compilation_sealed).

$$\Delta\, n = R \ \wedge\ \mathsf{NoRes}\ [\![R]\!] \ \wedge\ \mathsf{WfSig}\ n\ [\![R]\!]\ [\![\Gamma_1]\!] \ \wedge\ \Delta;\Gamma \vdash e_1 : \Gamma_1[n{:=}R] \rightsquigarrow \overline{e}_1 \ \wedge\ \Delta_c \sqsubseteq \Delta \wedge$$
$$\Delta_c;\Gamma \vdash e_2 : \mathsf{sig}\,(\{\ell : A\} \Rightarrow \langle B\rangle) \rightsquigarrow \overline{e}_2 \ \wedge\ \ell : A \in \Gamma_1 \ \wedge\ [\![\Gamma_1]\!] * [\![\Gamma]\!] \ \wedge\ [\![\Gamma_1]\!] * [\![B]\!] \wedge$$
$$\rho \vdash (\mathsf{seal}\ n\ R\ \Gamma_1\ e_1) \ltimes e_2 \Downarrow v \ \wedge\ \rho\ \mathsf{value} \ \wedge\ \Delta \vdash \rho \sim_\Gamma \overline{\rho}$$
$$\Longrightarrow\ \exists \overline{v}.\ \overline{\rho} \vdash \overline{s} \,\#\, \big((? : [\![\Gamma]\!]) \triangleright (\overline{e}_2\ \{\ell = \overline{s}.\ell\})\big) \hookrightarrow^* \overline{v} \ \wedge\ \Delta \vdash v \sim_{\Gamma_1 \& B} \overline{v} \qquad \textit{where } \overline{s} := \mathsf{seal}\ n\ [\![R]\!]\ [\![\Gamma_1]\!]\ \overline{e}_1$$

SFinitary c is the source-side normalizing fragment (every source former except the fixpoints); elaboration maps it into Finitary (elabSeal_finitary).

Theorem 2.10 (source_representation_independence).

$$\mathsf{OpenBrand}\ [\![\Delta_1]\!]\ [\![\Delta_2]\!]\ \eta\ n\ [\![R_1]\!]\ [\![R_2]\!] \ \wedge\ \Delta_1\, n = R_1 \ \wedge\ \Delta_2\, n = R_2 \wedge$$
$$\mathsf{WfSig}\ n\ [\![R_1]\!]\ [\![S]\!] \ \wedge\ \mathsf{WfSig}\ n\ [\![R_2]\!]\ [\![S]\!] \ \wedge\ \Delta_1;\varepsilon \vdash p_1 : S[n{:=}R_1] \rightsquigarrow \overline{p}_1 \ \wedge\ \Delta_2;\varepsilon \vdash p_2 : S[n{:=}R_2] \rightsquigarrow \overline{p}_2 \wedge$$
$$\mathcal{V}[\![[\![S]\!]]\!]^{(n,[\![R_1]\!],[\![R_2]\!])}(\overline{p}_1, \overline{p}_2) \ \wedge\ \emptyset; S \vdash c : \mathsf{Int} \rightsquigarrow \overline{c} \ \wedge\ \mathsf{SFinitary}\ c \wedge$$
$$\varepsilon \vdash (\mathsf{seal}\ n\ R_1\ S\ p_1) \triangleright c \Downarrow v_1 \ \wedge\ \varepsilon \vdash (\mathsf{seal}\ n\ R_2\ S\ p_2) \triangleright c \Downarrow v_2$$
$$\Longrightarrow\ \exists i.\ v_1 = i \ \wedge\ v_2 = i$$

3 Ongoing work

Planned results are typeset in red; statements are provisional (the source syntax for generics is not yet fixed — a brand-binding functor$^\beta$ or a separate bfunctor form).

Theorem 3.1 (source_generic_elaboration (planned)). *Source-side generics elaborate to the target $\forall\beta$ layer and are semantically preserved: extending SCE with a brand-binding functor form* bfunctor$(\beta).\,e$ *elaborated to* $\Lambda_{[\![A]\!]}.\,\overline{e}$ *and link-time instantiation elaborated through* $\cdot\,[\alpha_n]$*, the elaboration type-preservation and simulation theorems (seal_type_preservation, seal_semantic_preservation) extend to the new forms.*

Theorem 3.2 (seal_separate_compilation_generic (planned)). *Separate compilation for generic clients across brand* renaming*: a generic client compiled against an abstract signature links with a provider whose brand was allocated differently — renaming a concrete brand α_n to a fresh $\alpha_{n'}$ in store, derivation and program preserves typing and evaluation (freshness is genuinely required here, unlike for $\forall\beta$-instantiation).*

Theorem 3.3 (source_generic_client_representation_independence (planned)). *The source-level mirror of generic_client_representation_independence: a source generic client, instantiated at the one brand both source providers seal at (two representations, two stores, as in the target theorem), computes the same literal for providers related at the signature — by semantic preservation, the target theorem, and source determinism. The cross-brand version — one client, providers allocated at* different *brands — factors through seal_separate_compilation_generic and is not claimed here.*

The concrete work items, in dependency order:

- **Milestone 5 — source-side generics.** Add the brand-binding functor form to SCE, its elabSeal rules targeting $\Lambda/\cdot\,[\alpha_n]$, and the Correctness cases; instantiation happens at link time ($\ltimes^*$).
- **Brand-renaming metatheory.** Renaming concrete brand $\to$ concrete brand with freshness side conditions, through stores, derivations and values; culminates in seal_separate_compilation_generic. A separate mechanism from $\forall\beta$-instantiation, which is freshness-free by design.
- **Source-level generic example.** The $\{mk\}\,\&\,\{get\}$ example with a source $\forall$-functor client serving both sealed source providers — the source mirror of the mechanized target-level run.

- **μ/fix examples.** A sealed-interface factorial through the fixpoint closure, a tagged-union run, and a μ-stream, exercising the Phase-1 layers end-to-end in Examples.lean.
- **Step-indexed logical relation.** Lift the Finitary/SFinitary fence so the relational results (and RI) cover fixpoints and a structural μ-clause; replaces the termination-flavored fundamental lemma with a step-indexed one.
- **Lifting the $\forall$-fences.** A shifting μ-substitution admits $\forall$ inside μ-bodies ($\neg\forall \in T$ on fold/unfold); shifting in the term-level instantiation admits nested, multi-binder generics (BClosed 1 A on TBAPP).
- **Relational parametricity proper.** Free theorems over arbitrary admissible relations at β need brand-interpretation environments plus a properly-bound (EBClosed) typing discipline; representation independence of generic clients deliberately avoided both.
- **Cleanups.** Collapse the module-type syntax into the function type, merge single-import linking into the n-ary form, and update this document as the source syntax for generics lands.